# Outbound network design packet — how the job is done, step by step

**You.** Rajesh Rishiyur Chandrasekaran. Fifteen years making a network's plan match what moves. Career facts in this packet are reconciled to the Amazon TESS team-match packet (November 7, 2026) and the Apple CapEx call packet v3 (November 5, 2026); every career number carries the status those packets gave it.

**The job this packet prepares you for.** Own the strategy and design of a large-scale outbound transportation and sortation network. Design and roll out new network models; prove their cost and speed against the current network; build the profitability model for sortation capacity by region and business line; write and defend capital plans; design and run dispatch pilots; define and report transportation cost per package and performance against plan; turn models into plans the field can run; write recommendations for senior leadership; mentor on modeling standards. Each responsibility is worked in Part 5.

**What this packet is.** A separate, self-contained booklet. Part 1 gives the outbound machine as stocks and flows. Parts 2 and 3 walk what happens from the moment an order is placed, inside the fulfillment center, on the trucks, through the sort center and delivery station, to the door, with the unit of work, the rate, the capacity, the cost, and the metric at every step. Part 4 walks one package and one 100,000-package day end to end. Part 5 is how to do each line of the job, with the math. Part 6 builds the week-over-week scorecard and ranks the metrics. Part 7 maps your record onto the job. Part 8 is the formula sheet, the inputs card, the numbers card, and the public facts.

**Honesty labels.** PUBLIC AMAZON PRACTICE a mechanism Amazon has described in public, cited in Part 8 · INDUSTRY STANDARD textbook operations and supply-chain method · ILLUSTRATIVE a number chosen to make the arithmetic legible; never Amazon's figure · RECORD a reconciled career fact · USER ANCHOR carried in an earlier packet, confirm from memory · **model** your own arithmetic or design. No internal system name appears anywhere in this packet. Where a generic name is needed, it is a generic name: the promise engine, the ship sorter, the transportation plan.

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## How to use it

| PASS | WHAT TO DO | WHAT YOU SHOULD BE ABLE TO DO AFTERWARDS |
|---|---|---|

| | | |
|---|---|---|
| First read | Read the sixty-second answer and Part 1 until you can redraw the stock diagram from memory with the sixteen stocks in order. | Say what a stock, a flow, a capacity, and a cutoff are, and point to where each lives on the path from order to door. |
| Second read | Parts 2 and 3. Recompute every math block from the row above it. The reference day is one fulfillment center shipping 100,000 packages; every number in the packet ties to it. | Do the pick-hours, trailer-fill, sort-window, and route-density arithmetic out loud, and say what breaks when each number is wrong. |
| Third read | Part 4. Walk the one package and the one day without the page. Then Part 5, one responsibility at a time, saying the steps and the math. | Answer "how would you decide whether a sort center earns its cost" or "how would you pilot a dispatch change" with the inputs, the model, the decision rule, and the metric that proves it. |
| Night before | Part 6 (the scorecard), Part 7 (your record), Part 8 (the formula sheet and the numbers card). | Name the four output metrics and the eight inputs, rank them, say why, and tie each to a story you actually ran, with its label. |

**Two discipline rules for the room.** First, every number in this packet's worked examples is illustrative unless it carries a public source in Part 8 or a career status in the numbers card. Say "in a reference network I built, with the assumptions stated" and not "Amazon's cost per package is." Second, the career numbers keep their nouns: the Elementum lane release (about $1.5 million, one time), the Elementum forecast release (about $2 million, one time), the Elementum onboarded-lane freight saving (about $2 million in six months, user anchor), and the Flexport roadmap avoidance (about $5 million, said once) are four different sentences. None of them is added to another.

## The sixty-second answer — what outbound is, as stock and flow

Say this version out loud, then stop. It is the whole packet in one paragraph.

> **SAY THIS — 183 WORDS**
>
> Outbound is a chain of stocks and flows running against a clock. An order becomes a promise, the promise becomes a critical pull time, and from then on the package moves through a series of queues: pick, pack, label, sort, the dock, the trailer, the sort center, the second trailer, the delivery station, the van. Each queue is a stock; each process is a flow with a rate and a capacity; each handoff has a cutoff. The package makes the promise when every stock empties ahead of its cutoff, and it costs what each resource cost divided by how many packages rode it. So the design job has three objects. The clock: set the pull times from the measured transit tail, not the average. The denominators: fill on the trailer, packages on the route, volume through the sorter. And the plan: a weekly scorecard that reads cost per package and on-time against plan, decomposed into the inputs a building can move this week. Everything else is which node to build, which lane to run, and how to prove it before you scale it.

The one mental model to keep on top. **Outbound is a reservation system running against a promise.** The promise is made first, from inventory position and a transit table. Capacity is then reserved behind it: labor hours, sorter throughput, trailer positions, van routes. Most outbound failures are a stock that did not empty before its cutoff, and most outbound cost is a denominator that was smaller than the plan assumed. The metrics in Part 6 exist to catch those two things one week earlier than the customer does.

```
THE CHAIN IN ONE LINE                                                    [industry standard framing; node

 order --> promise --> pull time --> [pick] --> [pack] --> [SLAM] --> [ship sort] --> [dock] --> trailer
   --> linehaul --> [sort center window] --> trailer --> linehaul --> [delivery station sort] --> [stage]

 [ ] = a stock that must empty before the next cutoff.   --> = a flow with a rate and a capacity.
 Cost per package = sum over the path of (resource cost / packages that rode it).
 Promise kept = every stock on the path empty before its cutoff, with slack left for the tail.
```

# Part 1 — The outbound machine as stocks and flows

## 1.1 The vocabulary and the eight equations

A stock-and-flow model has three kinds of object. A **stock** is a count of things sitting or moving at one point: packages waiting to be packed, trailers in the yard, packages on vans. A **flow** is a rate into or out of a stock: packages packed per hour, trailers departed per cutoff. A **cutoff** is a vertical line on the clock that a stock must be empty by. The whole outbound network is sixteen stocks in series, each drained by a process with a capacity, each with a cutoff inherited from the one after it. The eight equations below are the only arithmetic the packet uses; everything in Parts 2 to 6 is one of them substituted with the reference day's numbers.
INDUSTRY STANDARD for the equations; ILLUSTRATIVE for every substitution.

**The eight equations, substituted on the reference day**

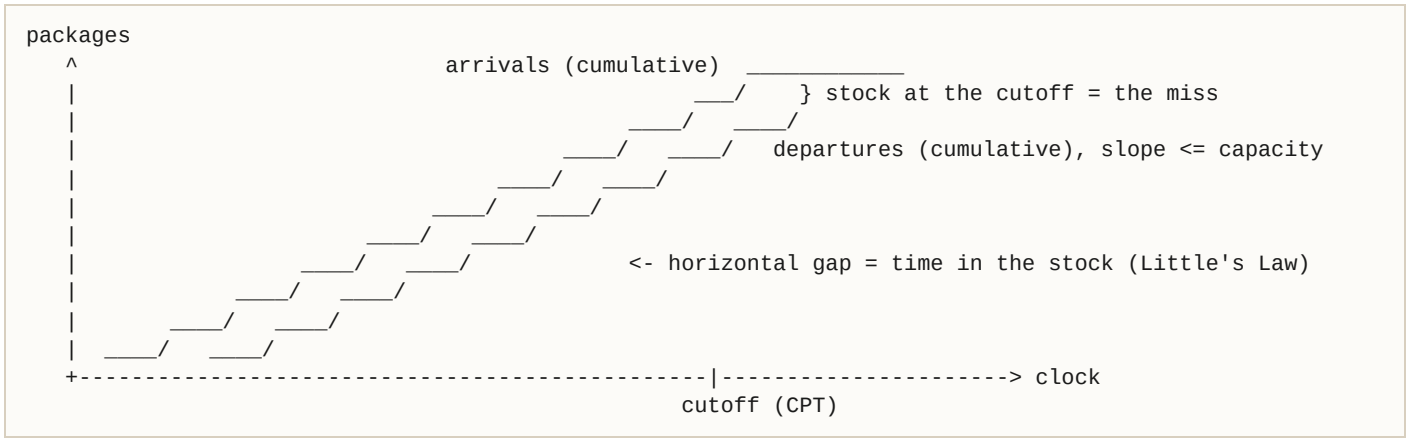
| # | EQUATION | SYMBOLIC | SUBSTITUTION ON THE REFERENCE DAY | WHAT IT IS FOR |
|---|---|---|---|---|
| 1 | Stock balance | $S(t+1) = S(t) + In(t) - Out(t)$ | Pack backlog at 13:00 = 500 + 5,000 in − 5,000 out = 500 (table 2.5) | Every queue in the building, every hour. The CPT miss is S at the cutoff. |
| 2 | Capacity | C = resources × rate × hours | SLAM: 3 lines × 2,500 per hour = 7,500 per hour. Pick: 58 pickers × 160 units per hour = 9,280 units per hour per shift | The most a flow can be. Demand above it becomes stock. |
| 3 | Utilization | ρ = demand ÷ capacity | SLAM average 5,000 ÷ 7,500 = 66.7%; peak hour 7,000 ÷ 7,500 = 93.3% | Where the queue starts to grow faster than the volume. |
| 4 | Little's Law | time in stock = stock ÷ throughput | 2,500 packages in the pack queue ÷ 5,000 per hour = 0.5 h = 30 minutes of wait | Turns a count you can see into a delay you can promise against. |
| 5 | Queue wait (Kingman, qualitative) | $W_q \approx [\rho \div (1-\rho)] \times [(c_a^2 + c_s^2) \div 2] \times t_s$ | Doors at 70%: factor 2.33; at 85%: 5.67; at 95%: 19.0. The last ten points triple the wait. | Why dock doors, inducts, and sorters are planned with slack near 85%. |
| 6 | Pipeline delay | in-transit stock = departure rate × transit time | FC-A to SC-1: 28 trailers a day ÷ 24 h × 4.0 h = 4.7 trailers on the road at any moment | Transit is a stock too. It is why a lane needs more trailers than departures. |
| 7 | Cutoff cascade | $cutoff_n = cutoff_{n+1} - transit - processing - buffer$ | Route 09:00 Tue → station cutoff 06:30 → SC departure 04:00 → SC cutoff 01:00 → CPT 21:00 Mon → pack complete 19:00 Mon (table 3.6) | The critical pull time. Every upstream clock is inherited from the delivery window. |
| 8 | Cost per package | CPP = Σ (resource cost ÷ packages that rode it) | $640 trailer ÷ 1,600 packages = $0.40; $264 route ÷ 200 packages = $1.32; $13,000 fixed ÷ 100,000 sorted = $0.13 | The denominators are the levers: fill, density, volume through fixed capacity. |

Two habits make these equations usable in a room. Say the **denominator** before the rate, because every cost-per-package argument is really an argument about what rode the resource. And say the **quantile** before the

mean, because every cutoff is a promise against the tail of a distribution, and an average transit time is late about half the time.

### Cumulative curves: the picture under every cutoff

The cleanest way to see a stock against a cutoff is two cumulative curves. The upper curve is cumulative arrivals into the stock; the lower curve is cumulative departures out of it. The vertical gap between them is the stock at that moment (equation 1). The horizontal gap is the time a package spends in it (equation 4). A cutoff is a vertical line; the vertical gap at that line is the miss. Capacity is the maximum slope of the lower curve (equation 2). Table 2.5 is exactly this picture, in numbers, for one critical pull time.




## 1.2 The physical flow, order to door, with every stock numbered

The node names are the ones Amazon uses in public: fulfillment center, sort center, delivery station, and SLAM for scan, label, apply, manifest. PUBLIC AMAZON PRACTICE for the shape of the path (Part 8, Amazon Science 2022 and 2023, the fulfillment-center tour, the 2025 INFORMS paper). INDUSTRY STANDARD for the stock-and-flow framing. The sixteen stocks are this packet's numbering, not Amazon's.

```
THE PHYSICAL FLOW -- what the package does, with every stock numbered          [shape: public Amazon practice; nur

   customer order
        |  promise = ship date + transit (business days); an "order within" countdown while the clock is open
        v
 [S1 ] orders promised, awaiting assignment to a building ..... minutes       --> assignment: which FC, which ship
        v
 [S2 ] orders assigned, awaiting release to pick ............... minutes-hours --> release by CPT priority; batchi
        v
 [S3 ] pick backlog (units) ..................................... hours         --> PICK  (pickers x units/hour)
        v
 [S4 ] picked units in totes, awaiting pack / rebin ............ minutes       --> conveyance; multi-item orders c
        v
 [S5 ] pack queue (packages) ................................... minutes-hours --> PACK  (packers x packages/hour,
        v
 [S6 ] packed, awaiting SLAM ................................... minutes       --> SLAM  (weigh, verify, label, ma
        v
 [S7 ] labeled, on the ship sorter ............................. minutes       --> SHIP SORT to a door chute by la
        v
 [S8 ] staged at the door: the CPT pile ....................... 0-6 hours     --> LOAD into gaylords / fluid loac
        v
 [S9 ] trailer in the yard / gate out .......................... minutes       --> DEPART at the critical pull tir
        v
 [S10] in transit, FC -> sort center ........................... hours         --> LINEHAUL (miles, transit distri
        v
 [S11] at the sort-center door, awaiting unload ................ 0-2 hours     --> UNLOAD (doors x crew x rate)
        v
 [S12] sort-center WIP: induct -> sort -> containerize by station  <= window   --> SORT WINDOW closes; late packag
        v
 [S13] in transit, sort center -> delivery station ............. hours         --> SECOND LINEHAUL; station cutoff
        v
 [S14] delivery-station WIP: induct -> sort to route -> stage ... <= wave      --> ROUTE BUILD; vans loaded in the
        v
 [S15] on the van .............................................. route hours   --> DELIVERY (stops per hour inside
        v
 [S16] delivered: attempt on or before the promised date  ...... sink          --> the output metric

   Exception stocks (3.7): missorted at S12/S14; rolled at S8/S12/S14; late at S10/S13; damaged anywhere; undelive
   Bypasses: direct FC -> delivery station (skips S11-S13); same-day building (S3..S8 then straight to S14, one si
             carrier handoff at the FC dock (S9 to a carrier's network; cost is the carrier's rate).
```

Three things to notice before the detail. First, the stocks before SLAM are measured in *units and orders*; after SLAM they are *packages*. The label is where the object changes, and it is where the package is committed to a lane and a departure. Second, the two transit stocks, S10 and S13, are pipeline delays: nothing can be done to a package while it is in them, so the only lever is when it entered. Third, every stock from S8 onward has a hard cutoff attached: a trailer departure, a sort window, a route wave. Those cutoffs are the critical pull time chain in 3.6, and they are the configuration this job owns.

---

## 1.3 The information flow: plans travel ahead of the packages

Every physical move above is preceded by an information move. The plan is made at several horizons, and the scorecard in Part 6 is where the plan and the actuals meet. INDUSTRY STANDARD for the planning horizons; the sign-off points are this packet's design.

```
INFORMATION FLOW -- plans travel ahead of the packages

 demand forecast: packages by origin FC x destination (station or ZIP cluster) x ship method x week
        |
        v
 network plan: which lanes run, hubs vs direct, sort-center assignment, station catchments
        |
        v
 transportation plan: trailers by lane x departure (CPT), carrier vs own, planned fill
        |
        v
 labor plans: FC outbound hours, sort-center hours per window, station sort hours, routes per wave
        |
        v
 configuration: customer cutoffs, CPTs per lane, sort codes -> chutes, station ZIP catchments, holds
        |
        v
 execution: scans at every stock boundary (pick, pack, SLAM, load, depart, arrive, induct, sort, stage, o
        |
        v
 actuals vs plan --> variance --> scorecard (Part 6) --> re-plan, re-configure, or escalate
        ^                                                        |
        +------------------ the loop that keeps the plan honest -------+
```

The habit that matters: score a plan at the horizon it was made. A trailer plan built at T-7 days against a forecast is compared to the actual at T-0, and the error is attributed to the forecast or to the execution, not blended. That is the same discipline as scoring an arrival estimate frozen thirty days out against the actual at time zero, which is the Flexport comparison in your record (Part 7). It is also why the scorecard carries forecast accuracy as an input metric: every other plan number inherits its error.

---

## 1.4 Node by node: what happens, the unit of work, the rate, the metric, what breaks

This is the whole path in one table. Every rate is the reference day's ILLUSTRATIVE input, and every one is used again in Parts 2 to 4 where the arithmetic is worked. Read the "what breaks" column as the failure the metric exists to catch.

| STAGE (STOCK) | WHAT HAPPENS | UNIT OF WORK | RATE AND CAPACITY ON THE REFERENCE DAY | METRIC THAT WATCHES IT | WHAT BREAKS IF IT IS WRONG |
|---|---|---|---|---|---|
| Order and promise (S1) | The site shows a delivery date built from the ship date plus transit; an order is placed against it. | An order; a promise | Software; not a labor rate. 100,000 packages a day from this building. | Promise offered vs promise the path can make; share on each speed tier. | A date the path cannot make, or a slower date than the path could make. Both are configuration defects. |
| Assignment (S1 → S2) | The order is assigned to a fulfillment center that holds the units, has capacity, and can reach the station by the pull time at the lowest cost; a ship method and a CPT are attached. | An assignment | Software. The reference FC gets its 100,000 from its region. | In-region fulfillment share; cost per assignment vs the cheapest feasible. | A long ship from the wrong region: more miles, more touches, a later promise. |

| | | | | | |
|---|---|---|---|---|---|
| Release and pick (S2, S3) | Orders are released to pick in CPT priority and batched; units are picked into totes. | A unit picked | 160,000 units; 160 units per hour; 1,000 direct hours; 58 pickers per 10-hour shift. | Units per hour vs standard; pick backlog in hours; CPT-priority compliance. | A backlog that outlives the CPT; picking tomorrow's work while today's waits. |
| Consolidate and pack (S4, S5) | Multi-item orders are consolidated; each order is packed in a box or bag with dunnage. | A package packed | 100,000 packages; 100 per hour; 1,000 direct hours; 58 packers per shift. | Packages per hour; pack-complete time vs CPT minus the dock buffer. | Pack complete after the buffer: a package that misses a trailer that was itself on time. |
| SLAM (S6) | The package is scanned, weighed against the expected weight of its contents, labeled, and manifested. The label commits it to a lane and a departure. | A package labeled | 3 lines × 2,500 per hour = 7,500 per hour; average 5,000; peak hour 7,000 (93%). | Weight-mismatch rate; SLAM throughput vs peak hour; label-to-manifest completeness. | A mis-pick that passes the weight check becomes a customer defect; a SLAM queue at the peak hour becomes a CPT miss. |
| Ship sort and dock (S7, S8) | The sorter reads the label and sends the package to the chute for its lane and CPT; loaders fill gaylords or fluid-load the trailer at the door. | A package sorted; a package loaded | Sorter 10,000 per hour (50% average, 70% peak). Load 500 per hour per loader; 200 direct hours. | Sort-to-chute accuracy; packages staged at CPT (the pile); fill at close. | A mis-sort rides to the wrong building. A pile at CPT is a roll. A light trailer is cost. |
| Trailer and departure (S9) | The trailer closes at the critical pull time, is sealed, pulled from the door, and leaves the gate. | A trailer departed | 2,000 positions; planned fill 80% = 1,600; 54 own trailers a day plus 6 carrier pickups. | CPT compliance (departed on time); fill at close; yard dwell. | A late departure spends slack the sort center does not have. An early departure leaves the pile behind. |
| Linehaul to the sort center (S10) | The trailer runs the lane. Transit is a distribution, not a number. | A trailer-mile; a trailer-trip | 200 miles at $3.20 per mile = $640; mean 4.0 h; p90 4.45 h; p99 4.8 h. | On-time arrival against the sort-center window; cost per trailer; cost per package = $640 ÷ 1,600 = $0.40. | A late trailer arrives after its window and its packages wait three hours for the next, or a day. |
| Sort-center unload and induct (S11, S12) | Trailers hit doors in the window; crews unload; packages are inducted (scanned) onto the sorter. | A trailer unloaded; a package inducted | 12 doors; 2 unloaders × 400 per hour = 2 h per trailer; 15.6 trailers per 3-h window; 10 induct stations at 1,000 per hour. | Door utilization and dwell; induct rate vs window; trailers past the window. | Bunched arrivals queue at the doors; the last trailers finish after the window and roll. |
| Sort and containerize (S12) | The sorter sends each package to the chute for its delivery station; chutes are built into bags or gaylords labeled to the station; containers are loaded to station trailers. | A package sorted; a container built | Sorter 20,000 per hour (42%); containerize and load at 600 per hour each; 14 heads each per window. | Sort completion vs window; missort rate (0.3% = 300 a day); containers per trailer. | A missort is a second sort, a second linehaul, and often a lost day. A window that runs long makes the FC's CPT fiction. |

| | | | | | |
|---|---|---|---|---|---|
| Linehaul to the station (S13) | Station trailers depart at the sort-center cutoff and run to the delivery station. | A trailer-trip | 120 miles, $384, 2.5 h; $384 ÷ 1,600 = $0.24 per package. | On-time arrival against the station cutoff; fill. | Arrival after the station's inbound cutoff: the package misses the wave. |
| Delivery-station sort and stage (S14) | Packages are unloaded, inducted, sorted to a route in stop sequence, and staged in bags or carts by route; vans are loaded in the wave. | A package sorted to route; a route staged | 6,000 packages; 30 routes of 200; sort 250 per hour; 5 sorters in a 5-h window; stage 400 per hour. | Sort completion vs route start; rolled volume; routes planned vs dispatched. | Overflow: more packages than routes, vans, or drivers. The excess rolls or goes out at a higher cost per package. |
| Route and door (S15, S16) | The driver runs the route inside the delivery window and makes the attempt at each stop; proof of delivery; exceptions come back to the station. | A stop; a delivery attempt | 160 stops per route; 20 stops per hour; 8-h route; $264 per route = $1.32 per package, $1.65 per stop. | Attempts on or before the promised date; stops per hour; packages per route; undeliverable rate. | Thin routes cost more per package; long routes miss the window. Both are catchment and wave decisions. |

**SAY THIS — 92 WORDS**

From the order to the door there are sixteen stocks and about a dozen processes. Before the label the work is units and orders; after it the work is packages committed to a lane. Each process has a rate and a capacity, each handoff has a cutoff, and each cutoff is inherited from the delivery window backward. The job is to keep every stock emptying ahead of its cutoff with slack for the tail, and to keep the denominators, fill, route size, and volume through fixed capacity, where the plan assumed them.

# Part 2 — Inside the fulfillment center: order placed to trailer closed

The reference building is FC-A: 100,000 packages a day, 1.6 units per package, two ten-hour shifts, four critical pull times a day on each sort-center lane. Every input is ILLUSTRATIVE and is listed again in 8.2. The sequence of steps is the public one: pick, pack, SLAM, sort to the trailer PUBLIC AMAZON PRACTICE. The rates, the clocks, and the dollars are the generic model.

## 2.1 Order, promise, and assignment

**What happens.** The customer sees a delivery date. Amazon's help page describes that date as the shipping date plus transit time in business days, with an "order within" countdown while the clock is open PUBLIC AMAZON PRACTICE. Behind the date are three decisions that happen in seconds: which building ships it, by which ship method, and against which critical pull time. The building is chosen from the ones that hold the units, have outbound capacity in the window, and can reach the customer's delivery station by its cutoff, at the lowest cost. The CPT attached to the order is the latest departure from that building that still makes the promised date. That is the clock everything in the building runs to.

**Customer cutoff, from the CPT backward**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | CPT on the SC-1 lane for next-day to the DS-7 catchment | From the chain in 3.6: route 09:00 Tue minus the four path legs | 21:00 Monday |
| 2 | Dock buffer | Pack complete must lead the trailer close by 2.0 h (sort, stage, load, seal) | Pack complete by 19:00 |
| 3 | Order-to-pack cycle at the 90th percentile | Pick queue 0.5 + pick 0.3 + convey 0.2 + pack queue 0.5 + pack and SLAM 0.2 + sort and stage 0.3 | 2.0 h at p90 (the mean is about 1.2 h) |
| 4 | Customer cutoff for next-day on this path | 21:00 − 2.0 − 2.0 | 17:00 Monday. The "order within" countdown ends here. |
| 5 | What a 1-hour slower building does to the cutoff | If the p90 cycle is 3.0 h: 21:00 − 2.0 − 3.0 | 16:00. One hour of building cycle time is one hour of customer cutoff, which is a share of next-day orders lost. |

**Assignment: two buildings that both hold the unit**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | FC-A, in region, via SC-1 to DS-7 | Dock-to-door from 2.7 and 3.5: linehaul 0.40 + sort 0.30 + linehaul 0.24 + station 0.30 + route 1.32 | $2.56; next-day feasible (CPT 21:00) |
| 2 | FC-B, out of region, 600 miles to a hub then to SC-1 | 600 × $3.20 ÷ 1,600 = $1.20 linehaul + hub sort 0.30 + 0.40 + 0.30 + 0.24 + 0.30 + 1.32 | $4.06; adds a hub touch and a day: two-day promise |
| 3 | Assignment rule | Lowest cost among feasible buildings; feasibility = units on hand, capacity in the window, pull time reachable | FC-A. FC-B is the fallback when FC-A is out of stock or out of capacity. |

| | | | |
|---|---|---|---|
| 4 | What the fallback costs when it fires | $4.06 − $2.56 | $1.50 per package and one promise tier. The public version of this is in-region fulfillment going from 62 to 76 percent after regionalization (Part 8). |

**What the assignment is not.** It is not a choice the building makes. The building receives the order with its CPT already attached. The design job sits upstream: what the transit table says, which lanes exist, where the inventory was placed. A building that misses pull times it was never able to make is a configuration defect, not a building defect.

## 2.2 Release and pick

**What happens.** Assigned orders wait in S2 until they are released to the floor. Release is by CPT priority: the orders whose pull time is soonest go first, and orders for later departures fill idle capacity. Released work is batched into pick tasks so that one picker's tour serves many orders. In a pod-to-person building the pods come to the picker; in a manual building the picker walks the path; either way the unit of work is a unit picked into a tote, and the rate is units per hour. The tote rides conveyance to consolidation or pack. The stock that matters is the pick backlog in hours, not units: units ÷ (pickers × rate).

**Pick labor on the reference day, and the learning-curve tax**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Units to pick | 100,000 packages × 1.6 units per package | 160,000 units |
| 2 | Direct pick hours | 160,000 ÷ 160 units per hour | 1,000 direct hours |
| 3 | Paid hours | 1,000 × 1.15 (15% indirect: breaks, meetings, problem-solve) | 1,150 paid hours |
| 4 | Heads per shift | 1,150 ÷ 2 shifts ÷ 10 hours | 57.5, roster 58 |
| 5 | Pick capacity per hour on a shift | 58 × 160 | 9,280 units per hour = 5,800 packages per hour at 1.6 |
| 6 | Backlog in hours when 12,000 units are waiting | 12,000 ÷ 9,280 (Little's Law) | 1.29 h. Say the backlog in hours; units hide the clock. |
| 7 | Peak with 30% new hires at 60% of standard | 0.7 × 160 + 0.3 × 96 = 112 + 28.8 | 140.8 units per hour blended |
| 8 | Hours at the blended rate | 160,000 ÷ 140.8 = 1,136.4; 1,136.4 − 1,000 | +136 direct hours, +13.6%, paid before the first peak trailer |

**The priority rule, in stock terms.** If the floor picks in arrival order, the pick backlog for the 21:00 CPT grows while pickers finish two-day orders that could wait until tomorrow. If it picks in CPT order, the urgent backlog drains first and the flexible work absorbs the troughs. Table 2.5 is built on that rule: urgent packages are served first, flexible work fills whatever capacity is left.

## 2.3 Pack and SLAM: the label is the commitment

**What happens.** Single-unit orders go straight to pack. Multi-unit orders are consolidated first, because their units were picked by different people into different totes. The packer selects the box or bag, adds dunnage, and closes it. The package then goes to SLAM: it is scanned, weighed, the weight is compared with the expected weight of its contents, the shipping label is printed and applied, and the package is manifested to its ship method and departure PUBLIC AMAZON PRACTICE for the SLAM step and its position between pack and the outbound trailer.

Two things happen at SLAM that matter to the network. The weight check is the last automatic chance to catch a mis-pick before it leaves the building. And the label fixes the lane and the departure: after this point, changing the package's path is a rehandle.

**Pack and SLAM capacity against the peak hour**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Direct pack hours | 100,000 ÷ 100 packages per hour (blended single and multi) | 1,000 direct, 1,150 paid, 58 per shift |
| 2 | SLAM capacity | 3 lines × 2,500 per hour | 7,500 per hour |
| 3 | Average and peak demand | 100,000 ÷ 20 h = 5,000; peak-hour factor 1.4 | 5,000 average; 7,000 at the peak hour |
| 4 | Utilization | 5,000 ÷ 7,500; 7,000 ÷ 7,500 | 66.7% average; 93.3% at the peak hour |
| 5 | A 10-minute SLAM stop in the peak hour | Arrivals keep coming: 7,000 × 10 ÷ 60 | 1,167 packages queued in front of SLAM |
| 6 | Time to drain the queue while arrivals continue | queue ÷ (capacity − arrival rate) = 1,167 ÷ (7,500 − 7,000) | 2.33 hours. At 93% utilization a 10-minute stop costs the rest of the peak. |
| 7 | Same stop at the average hour | 5,000 × 10 ÷ 60 = 833; 833 ÷ (7,500 − 5,000) | 20 minutes. Same stop, seven times faster recovery. |
| 8 | Weight-check catch | If 0.2% of packages fail the tolerance and 60% of those are real mis-picks | 200 kicked to problem-solve a day, 120 real. Illustrative; the point is that the check exists before the label. |

Step 6 is the stock-and-flow lesson that is hardest to see from a labor plan. Capacity minus arrival rate is the drain rate. Near full utilization the drain rate is tiny, and any stop becomes a pile that outlives the hour. That is why SLAM, the ship sorter, and the sort-center inducts are planned with slack at the peak hour, not at the average.

---

## 2.4 Ship sort, the dock, and the critical pull time pile

**What happens.** The labeled package rides the ship sorter, which reads the label and diverts it to a chute. A chute is a destination and a departure: the SC-1 lane's 21:00 trailer, the SC-2 lane's 21:00 trailer, direct DS-3's 20:00 truck, the carrier's 19:00 pickup. At the door, loaders build gaylords or fluid-load the trailer that is live for that chute. Packages that arrive at the chute before the trailer closes ride; packages that arrive after it closes become the pile for the next departure. The stock at the door at the moment of CPT is the single most important number in the building's day: it is the miss.

**Chutes, doors, and loaders**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Live outbound destinations | SC-1 (7 trailers per CPT), SC-2 (4), 5 direct stations (1 each), 3 carrier pickups | 19 live doors. At 24 outbound doors: 79% in use, 5 for swaps and recovery. |
| 2 | Ship sorter | Capacity 10,000 per hour; average 5,000; peak 7,000 | 50% average, 70% peak. The sorter is not the constraint; its chute count is. |

| | | | |
|---|---|---|---|
| 3 | Load labor | 100,000 ÷ 500 per hour per loader = 200 direct; × 1.15 | 230 paid hours; about 12 loaders per shift |
| 4 | Packages per gaylord and per trailer | 50 per gaylord; 40 gaylord positions (20 floor positions, double-stacked) | 2,000 packages at 100% of positions |
| 5 | Load time for one trailer at plan | 1,600 ÷ (2 loaders × 500) | 1.6 h of loading inside a 6-h accumulation window. The door is occupied by the window, not by the work. |
| 6 | What a chute shortage costs | If SC-1 and SC-2 share a chute, the sort center receives mixed trailers and must separate them | An extra sort touch downstream, about $0.13 of labor per package, and a slower window. Chute count is a design standard, not a floor detail. |

**Load to schedule or load to fill.** A trailer that closes at the CPT regardless of fill protects the clock and spends cost. A trailer that waits for fill protects cost and spends the clock. The right policy on a lane depends on the slack downstream, which is in 3.6, and on the fill distribution, which is in 2.6. The pilot in 5.4 is the controlled test of a rule that mixes the two.

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## 2.5 One CPT cycle, hour by hour: base, plus thirty, plus sixty

One departure group: the four CPTs at 15:00 across all lanes, 25,000 urgent packages. Orders for it arrive between the previous customer cutoff at 08:00 and this one at 14:00; pack must be complete by 13:00 for a 15:00 close. For legibility the table shifts the clock so the window runs 08:00 to 16:00: arrivals 08:00 to 14:00, pack complete by 16:00. Combined pick-pack-SLAM capacity is 5,000 packages per hour. Urgent work is served first; flexible work (later CPTs, two-day) takes whatever capacity is left. ILLUSTRATIVE throughout; the mechanism is equation 1.

**Base day: 25,000 urgent packages, 5.0 hours of work in an 8-hour window**

| HOUR | URGENT IN | AVAILABLE (BACKLOG + IN) | SERVED (≤ 5,000) | CAPACITY LEFT FOR FLEXIBLE WORK | URGENT BACKLOG AT HOUR END |
|---|---|---|---|---|---|
| 08–09 | 2,500 | 2,500 | 2,500 | 2,500 | 0 |
| 09–10 | 3,000 | 3,000 | 3,000 | 2,000 | 0 |
| 10–11 | 4,000 | 4,000 | 4,000 | 1,000 | 0 |
| 11–12 | 5,000 | 5,000 | 5,000 | 0 | 0 |
| 12–13 | 5,500 | 5,500 | 5,000 | 0 | 500 |
| 13–14 | 5,000 | 5,500 | 5,000 | 0 | 500 |
| 14–15 | 0 (cutoff passed) | 500 | 500 | 4,500 | 0 — pack complete 14:06 |

Slack to the 16:00 pack-complete line: 1.9 hours. Peak backlog 500 packages = 6 minutes of wait at 5,000 per hour. The flexible work absorbed 5,500 packages of capacity before noon and 4,500 more once the urgent work cleared.

**Plus thirty percent: 32,500 urgent packages, 6.5 hours of work**

| HOUR | URGENT IN | AVAILABLE | SERVED | LEFT FOR FLEXIBLE | BACKLOG |
|---|---|---|---|---|---|
| 08–09 | 3,250 | 3,250 | 3,250 | 1,750 | 0 |
| 09–10 | 3,900 | 3,900 | 3,900 | 1,100 | 0 |

| | | | | | |
|---|---|---|---|---|---|
| 10–11 | 5,200 | 5,200 | 5,000 | 0 | 200 |
| 11–12 | 6,500 | 6,700 | 5,000 | 0 | 1,700 |
| 12–13 | 7,150 | 8,850 | 5,000 | 0 | 3,850 |
| 13–14 | 6,500 | 10,350 | 5,000 | 0 | 5,350 |
| 14–15 | 0 | 5,350 | 5,000 | 0 | 350 |
| 15–16 | 0 | 350 | 350 | 4,650 | 0 — pack complete 15:04 |

Slack 0.93 h. Peak backlog 5,350 packages = 64 minutes of wait. The flexible work got nothing from 10:00 to 15:00, so the two-day orders are now also late in their own cycle.

**Plus sixty percent: 40,000 urgent packages, 8.0 hours of work in an 8-hour window**

| HOUR | URGENT IN | AVAILABLE | SERVED | LEFT FOR FLEXIBLE | BACKLOG |
|---|---|---|---|---|---|
| 08–09 | 4,000 | 4,000 | 4,000 | 1,000 | 0 |
| 09–10 | 4,800 | 4,800 | 4,800 | 200 | 0 |
| 10–11 | 6,400 | 6,400 | 5,000 | 0 | 1,400 |
| 11–12 | 8,000 | 9,400 | 5,000 | 0 | 4,400 |
| 12–13 | 8,800 | 13,200 | 5,000 | 0 | 8,200 |
| 13–14 | 8,000 | 16,200 | 5,000 | 0 | 11,200 |
| 14–15 | 0 | 11,200 | 5,000 | 0 | 6,200 |
| 15–16 | 0 | 6,200 | 5,000 | 0 | 1,200 at the pack-complete line |
| 16–17 | 0 | 1,200 | 1,200 | 3,800 | 0 — pack complete 16:14 |

The miss: 1,200 packages at 16:00, 4.8% of the CPT. The morning idle hours (1,200 packages of capacity in 08:00 to 10:00) cannot be moved to the afternoon; capacity is perishable. Options, each priced in 5.4: pull the customer cutoff 15 minutes earlier (loses orders), add 200 packages per hour of capacity from 10:00 (six hours × 200 = 1,200: two more packers and two more pickers on the shift), or let the 1,200 roll to the 21:00 CPT if that departure still makes the promise.

**SAY THIS — 130 WORDS**

A critical pull time is a stock that has to be zero at a line on the clock. I model it as cumulative arrivals against cumulative capacity, hour by hour. On the base day the backlog peaks at five hundred packages and pack is complete two hours early. At plus thirty the backlog peaks at five thousand three hundred and the slack is under an hour. At plus sixty the window and the work are both eight hours, and twelve hundred packages are still on the floor at the line. Capacity in the morning does not carry forward, so the fix is either the cutoff, the afternoon capacity, or a later departure that still makes the promise. I would price all three before I asked the building to run faster.

## 2.6 The trailer: positions, cube, weight, fill, and the roll risk

**What happens.** The trailer at the door is the unit the network pays for. Which constraint binds decides how many packages it carries: gaylord positions if the freight is containerized, usable cube if it is fluid-loaded, payload weight if the freight is dense. Small parcels almost never hit weight. The fill the plan assumes is a quantile choice: plan high and some days overflow; plan low and every day pays for air.

**Which constraint binds on a 53-foot trailer**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Positions | 20 floor positions × 2 high × 50 packages per gaylord | 2,000 packages at 100% of positions |
| 2 | Package cube at 100% of positions | 2,000 × 0.6 ft³ | 1,200 ft³ = 40% of 3,000 ft³ usable cube. The other 60% is air inside and between gaylords. |
| 3 | Fluid-load alternative | 3,000 × 0.85 packing ÷ 0.6 | 4,250 packages: 2.1 times the positions, at the cost of slower unload and no station-level separation inside the trailer |
| 4 | Weight at 100% of positions | 2,000 × 2.5 lb | 5,000 lb against a 45,000-lb payload. Weight does not bind; positions do. |
| 5 | Planned fill and planned packages | 80% × 2,000 | 1,600 per trailer. SC-1: 11,200 per CPT ÷ 1,600 = 7 trailers. |
| 6 | Linehaul per package at three fills | $640 ÷ (2,000 × 0.70); ÷ 1,600; ÷ 1,800 | $0.457 at 70%; $0.400 at 80%; $0.356 at 90% |

**Fill target as a quantile: the roll risk when daily volume has a 10% standard deviation**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Overflow probability at 80% plan | Mean 1,600, σ 160; z = (2,000 − 1,600) ÷ 160 = 2.50; 1 − Φ(2.50) | 0.6% of trailer-days overflow |
| 2 | At 85% plan | Mean 1,700, σ 170; z = 1.76 | 3.9% |
| 3 | At 90% plan | Mean 1,800, σ 180; z = 1.11 | 13.3% |
| 4 | Expected overflow packages per trailer | σ × [φ(z) − z × (1 − Φ(z))] | 0.3 at 80%; 2.7 at 85%; 12.1 at 90% |
| 5 | Trailer cost saved per 1,600 packages, versus the 80% plan | (1 − 1,600 ÷ planned packages) × $640 | $0 at 80%; $37.6 at 85%; $71.1 at 90% |
| 6 | Cost if overflow is cleared by an extra trailer | P(overflow) × $640 | $4.0 at 80%; $24.8 at 85%; $85.3 at 90%. At 90% the extra trailer eats the saving. |
| 7 | Cost if overflow rolls to the next departure | Expected overflow × $2.00 recovery per package | $0.6 at 80%; $5.3 at 85%; $24.3 at 90%. Cheaper, but 12 packages per trailer-day risk the promise. |
| 8 | The decision rule | Plan fill where saved trailer cost − expected roll cost is largest, subject to the roll not breaking the promise | If the next departure still makes the promise, plan toward 90% and roll. If it does not, plan near 80% and dispatch. The slack in 3.6 decides which lane is which. |

## 2.7 Fulfillment-center outbound cost per package, built up

The outbound handling cost is the labor and materials from release to the sealed trailer. Building, equipment, and inbound costs are not in it. This is the "$0.75 handling" line that the cost stacks in Part 3 use; here it is derived so you can defend it.

**FC outbound handling on the reference day**

| STEP | LINE | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Pick labor | 1,150 paid hours × $22 | $25,300 |
| 2 | Pack labor | 1,150 × $22 | $25,300 |
| 3 | Ship and dock labor | Loaders 230 + SLAM operators 60 + sorter operators 40 + problem-solve 100 = 430 paid hours × $22 | $9,460 |
| 4 | Labor per package | (25,300 + 25,300 + 9,460) ÷ 100,000 = 60,060 ÷ 100,000 | $0.60 |
| 5 | Packaging materials | Box or bag, dunnage, label; blended | $0.15 |
| 6 | Outbound handling per package | 0.60 + 0.15 | $0.75. Used as the handling line in every stack in Part 3. |
| 7 | Sensitivity: units per package 1.6 → 1.8 | Pick hours scale with units: 1,000 × 1.8 ÷ 1.6 = 1,125 direct; labor per package rises by 125 × 1.15 × 22 ÷ 100,000 | +$0.03 per package in the building, but fewer packages for the same units downstream. Units per package is a network lever, not a building lever: Amazon named higher units per package as a 2024 productivity driver (Part 8). |

**SAY THIS — 106 WORDS**

Inside the building the chain is pick, pack, label, sort, load. On a hundred-thousand-package day that is a thousand direct hours of pick, a thousand of pack, about four hundred of ship and dock, sixty cents of labor and fifteen cents of materials per package. The two clocks that matter are the peak hour at the label line, where ninety-three percent utilization turns a ten-minute stop into a two-hour queue, and the pile at the door at the critical pull time, which is the miss. The trailer carries two thousand positions; the plan fills eighty percent, because at ninety the overflow costs more than the air.

# Part 3 — The trucks, the sort center, the delivery station, the door

After the trailer closes, the package is in the network's hands: two linehauls, one sort center, one delivery station, one route. The node shapes are public PUBLIC AMAZON PRACTICE; every rate, clock, and dollar is the reference network ILLUSTRATIVE. The lane walked throughout is FC-A to SC-1 to DS-7, the path the Part 4 package rides.

## 3.1 Linehaul: miles, rate, the transit distribution, on-time arrival

**What happens.** A tractor pulls the sealed trailer from the yard at the CPT and runs the lane. The lane has a scheduled transit, but the truck has a distribution: traffic, weather, a yard queue at the far end. The sort center's window does not care about the schedule; it cares about the arrival. So the lane is planned on a quantile of the distribution, and on-time arrival is measured against the window the trailer had to make, not against the clock it was supposed to leave by.

**The three lanes out of FC-A**

| LANE | MILES | RATE PER MILE (ALL-IN) | COST PER TRAILER | PLANNED TRANSIT | PACKAGES PER TRAILER AT PLAN | LINEHAUL PER PACKAGE | DEPARTURES PER DAY |
|---|---|---|---|---|---|---|---|
| FC-A → SC-1 | 200 | $3.20 | $640 | 4.0 h | 1,600 | $0.40 | 28 (7 per CPT × 4) |
| FC-A → SC-2 | 150 | $3.20 | $480 | 3.0 h | 1,600 | $0.30 | 16 (4 × 4) |
| FC-A → DS-1..5 direct | 90 | $3.20 | $288 | 2.0 h | 1,500 (75%) | $0.19 | 10 (1 × 2 × 5 stations) |

The $3.20 per mile is a blended all-in: driver about $0.90, fuel $0.75, tractor $0.60, trailer $0.15, insurance, overhead, and an empty-mile allowance $0.80. If a lane has no backhaul, the empty return doubles the effective cost per loaded mile. Inbound transfers and returns are what fill the backhaul; a lane plan that ignores them overstates the cost of a direct lane and understates the cost of a hub lane. Illustrative build; a real rate card comes from the carrier contract or the fleet's cost model.

**Transit as a distribution, and what the pull time is planned on**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | FC-A → SC-1 transit, measured over a quarter | Mean 4.0 h, standard deviation 0.35 h, roughly normal on this lane (fat right tail on a weather lane) | p50 4.00 h; p90 = 4.0 + 1.28 × 0.35 = 4.45 h; p99 = 4.0 + 2.33 × 0.35 = 4.81 h |
| 2 | Departure at 21:00 against a 01:00 window open | Arrival = 21:00 + transit | p50 arrival 01:00; p90 01:27; p99 01:49 |
| 3 | What the sort center can absorb | Window 01:00–04:00; unload 2.0 h; sort and containerize the last packages 0.4 h | Latest arrival that still clears the window: 04:00 − 2.4 = 01:36. The window carries 36 minutes of arrival slack. |
| 4 | Share of trailers that miss the window at a 21:00 departure | P(transit > 4.6 h) = P(z > (4.6 − 4.0) ÷ 0.35 = 1.71) | 4.4% of trailers, about 1.2 of the 28 a day, 1,970 packages a day on average that roll a window |
| 5 | Option A: plan the departure on p90 | CPT = 01:36 − 4.45 = 21:09, round to 21:00 and keep the 36 minutes; or set the CPT at 20:45 and hold 51 minutes | At 20:45 the miss falls to P(z > (4.85 − 4.0) ÷ 0.35 = 2.43) = 0.8%; the customer cutoff moves 15 minutes earlier for the whole lane |

| | | | |
|---|---|---|---|
| 6 | Option B: a faster unload on the late trailer | Crew of 3 instead of 2: 1,600 ÷ 1,200 = 1.33 h; latest arrival = 04:00 − 1.73 = 02:16 | P(z > (5.27 − 4.0) ÷ 0.35 = 3.6) is under 0.1%. Same labor hours, more hands on the late door. Cheapest fix on this lane. |
| 7 | On-time arrival, the metric | Trailers arrived before (window close − unload − sort) ÷ trailers due in the window | Measured against the window, by lane. A trailer that left on time and arrived late is the lane's defect; a trailer that left late is the building's. Keep the two apart. |

**Equipment: transit is a stock**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Trailers on the road, FC-A → SC-1 | Departure rate × transit = 28 ÷ 24 × 4.0 | 4.7 trailers in transit at any moment |
| 2 | Round-trip cycle | 4.0 out + 2.0 at the sort-center door + 4.0 back + 1.0 at the FC yard | 11.0 h; a tractor can run about two round trips a day |
| 3 | Tractors for the lane | 28 departures ÷ 2 per tractor-day, plus 10% for maintenance and slack | About 15 tractors |
| 4 | Trailers in the pool | In cycle 28 × 11 ÷ 24 = 12.8; live at the FC doors 7; staged at the sort center 7 | About 27 trailers for one lane. The pool is why a lane change takes weeks, not a config edit. |

## 3.2 The sort center: doors, induct, sort, containerize, load, missort

**What happens.** SC-1 takes trailers from several fulfillment centers, FC-A among them, and sorts their packages by delivery station PUBLIC AMAZON PRACTICE for the role of the sort center; the 2025 INFORMS paper's hub and last-mile sort-center split is the designed version. In the reference network SC-1 is a last-mile sort center: it runs four three-hour windows a day; each window receives about 25,000 packages on about 16 trailers, inducts them onto a sorter, builds the chutes into bags and gaylords labeled to a station, and loads station trailers that leave at the window close. The window is the stock's cutoff. A package not in a station container when the window closes waits three hours for the next one, and whether that costs a day depends on the station's cutoff in 3.3.

**SC-1 capacity by step, one window of 25,000**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Packages and trailers per window | 100,000 ÷ 4; 25,000 ÷ 1,600 | 25,000 packages; 15.6, say 16 trailers (7 of them from FC-A) |
| 2 | Unload | 2 unloaders × 400 per hour = 800 per hour per door; 1,600 ÷ 800 | 2.0 h per trailer; 250 labor hours a day across the four windows |
| 3 | Door-slots per window | 12 doors × 3 h ÷ 2 h per trailer | 18 slots for 16 trailers: 87% door utilization, above the 85% knee, which is why arrival timing in 3.1 matters |
| 4 | Induct | 25,000 ÷ 3 h = 8,333 per hour; ÷ 1,000 per station ÷ 0.85 | 9.8, so 10 induct stations; 100 labor hours a day |
| 5 | Sorter | 8,333 per hour ÷ 20,000 capacity | 42%. Sized for the peak window, not the average one. |

| | | | |
|---|---|---|---|
| 6 | Containerize and load | 8,333 ÷ 600 per hour each | 14 heads building containers, 14 loading, per window; 167 labor hours a day each |
| 7 | Labor per day and per package | (250 + 100 + 167 + 167) = 683 direct × 1.15 = 786 paid × $22 = $17,288 ÷ 100,000 | $0.17 per package of labor |
| 8 | Fixed cost per package | Lease, sorter depreciation, utilities, salaried staff $13,000 a day ÷ 100,000 | $0.13. Total $0.30 per package sorted at 100,000 a day; $0.39 at 60,000; $0.27 at 130,000 if the sorter and windows allow. |
| 9 | Blended rate for a sort-center touch | 1 ÷ (1/400 + 1/1,000 + 1/600 + 1/600) | 146 packages per labor-hour, all four touches |

**The window as a queue: 16 trailers, 12 doors, 2 hours each**

| STEP | ARRIVALS | WHAT HAPPENS AT THE DOORS | RESULT |
|---|---|---|---|
| 1 | 6 trailers at 01:00 | 6 doors start; finish 03:00 | On time |
| 2 | 6 trailers at 01:30 | 6 more doors start; finish 03:30 | On time |
| 3 | 4 trailers at 02:00 | No free door until 03:00; start 03:00; finish 05:00 | One hour past the 04:00 close. 4 × 1,600 = 6,400 packages, 25.6% of the window, roll to the 07:00 window. |
| 4 | Why the average hid it | Door utilization for the window is 16 ÷ 18 = 89%; average dwell including the four is (12 × 2 + 4 × 3) ÷ 16 = 2.25 h | A 2.25-hour average looks fine. The marginal trailer waited an hour and lost the window. Report the tail, not the mean. |
| 5 | Fix 1: crew of 3 on every door | 1.33 h per trailer: 01:00 group done 02:20; 01:30 group done 02:50; 02:00 group starts 02:20, done 03:40 | Inside the window. Labor hours unchanged (250 a day), doors turned faster. Cost: zero on paper, a staffing change at the window peak. |
| 6 | Fix 2: stagger the FC departures | Move the 02:00 group's CPT 30 minutes earlier at their origin buildings | Inside the window. Cost: 30 minutes of customer cutoff on 4 trailers' worth of orders at those origins. |
| 7 | Fix 3: two more doors | 14 doors × 3 ÷ 2 = 21 slots; the 02:00 group starts on arrival | Inside the window. Cost: capital and yard space; the right answer only if the peak-window forecast says 16 becomes 20. |
| 8 | Fix 4: accept the roll | 6,400 packages take the 07:00 window and the 10:00 station departure | Inside the promise only if the station has an afternoon wave. In 3.5 that wave costs $1.90 per package instead of $1.32: 6,400 × $0.58 = $3,712 a day, every day the pattern repeats. |

**Missort**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Missorts a day at 0.3% | 100,000 × 0.003 | 300 packages ride to the wrong station |

| | | | |
|---|---|---|---|
| 2 | Cost per missort | Re-sort at the wrong station 0.30 + a second linehaul 0.24 + the share of customer contacts and concessions, blended | About $2.00 each, illustrative; $600 a day |
| 3 | Promise effect | If 70% of missorts lose the day: 300 × 0.7 ÷ 100,000 | 0.21 points of delivery estimate accuracy from one building's sort quality |
| 4 | The input behind it | Sort-code mapping (which station each ZIP belongs to), label quality, chute assignment, and the operator who closes a container with the wrong label | Missort is a configuration metric as much as a quality metric. The 2024 MIT thesis names missort as a large contributor to misses (Part 8). |

## 3.3 Sort center to delivery station: the second linehaul and the station cutoff

**What happens.** At the window close, the station containers are loaded and the station trailers leave. A station trailer is usually lighter than a sort-center trailer because a station is a smaller destination; a last-mile sort center that feeds many stations is trading the fill on this leg for the sort it saved upstream. The station has an inbound cutoff: the last arrival that can still be sorted into the morning wave.

**SC-1 → DS-7, and the station's actual cost on the leg**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Lane | 120 miles × $3.20; transit 2.5 h | $384 per trailer; departs 04:00, arrives 06:30, the station cutoff |
| 2 | Planned cost per package | $384 ÷ 1,600 | $0.24 |
| 3 | DS-7's actual inbound on the reference day | 3 trailers from SC-1 at 1,600 each = 4,800; 1 trailer from SC-2 at 1,200 (60% of positions) | 6,000 packages on 4 trailers |
| 4 | Actual cost per package on the leg | 4 × $384 ÷ 6,000 | $0.256, not $0.24. The light SC-2 trailer costs the station 1.6 cents on every package it receives. |
| 5 | The consolidation question | Should SC-2's 1,200 ride through SC-1 instead, adding a sort touch at $0.17 of labor and 45 minutes of clock? | $0.17 added to save $0.096 of linehaul on those 1,200 packages ($384 ÷ 1,200 − $0.24). No. Keep the light trailer; or re-draw DS-7's catchment so SC-2 sends it a full one. |

## 3.4 The delivery station: induct, sort to route, stage, load out

**What happens.** The station receives its trailers overnight, unloads, inducts each package, and sorts it to a route in stop sequence. Routes are built from the day's volume and the station's catchment; the packages for a route are staged in bags and carts by stop order; drivers check in, load out, and leave in the wave. The last mile is run by Delivery Service Partners, Amazon Flex drivers, and in some places local businesses PUBLIC AMAZON PRACTICE, with carriers such as the postal service and UPS still carrying part of the volume (Part 8). In the reference network DS-7 runs one morning wave of 30 DSP routes and an afternoon Flex wave for late arrivals and same-day volume.

**DS-7 on the reference day: 6,000 packages, 30 routes**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | Sort window and rate | 04:00–09:00, 5 h; 6,000 ÷ 5 = 1,200 per hour ÷ 250 per sorter | 4.8, so 5 sorters in the window |
| 2 | The last trailer | Arrives 06:30; unload 0.5 h; 1,600 ÷ (5 × 250) = 1.28 h to sort; stage 0.5 h | Staged by 08:47. Thirteen minutes of slack to the 09:00 wave. The station cutoff is real. |
| 3 | Labor | Unload 6,000 ÷ 600 = 10 h; sort 6,000 ÷ 250 = 24 h; stage and load-out 6,000 ÷ 400 = 15 h; 49 direct × 1.15 = 56.4 paid × $22 | $1,240 a day = $0.21 per package |
| 4 | Fixed | $560 a day ÷ 6,000 | $0.09; station handling $0.30 per package |
| 5 | Routes | 6,000 ÷ 200 packages per route | 30 routes, 30 vans, 30 drivers in the wave |
| 6 | Overflow: a plus-ten-percent day | 6,600 ÷ 200 = 33 routes; 3 more than planned, 600 packages | Three choices: 3 more DSP routes at $264 = $792 if vans and drivers exist by 09:00 (usually they do not); Flex blocks at $1.90 per package = $1,140; or roll 600 packages, 10% of the station's day, to tomorrow. |
| 7 | Rolled volume, the metric | Packages received for the wave that were not dispatched in it ÷ packages received | On this day 0% at plan; 9.1% if the overflow rolls. Reported by station, with the cause: no routes, late arrival, missort, damaged. |

## 3.5 The route and the door: stops, density, the window, the attempt

**What happens.** The driver runs the route in sequence inside the delivery window, makes an attempt at each stop, and records proof of delivery. Some stops fail: no safe location, access, a wrong address. Those packages come back to the station and are re-attempted or returned. The cost of the route is nearly fixed per route; what the package pays depends on how many packages shared the route and how many stops fit inside the window.

**Route economics and density**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Route cost | 8 h × ($22 driver + $8 van + $3 fuel and other) | $264 per route |
| 2 | Stops and rate | 200 packages ÷ 1.25 packages per stop = 160 stops; 160 ÷ 8 h | 20 stops per hour; the window 09:00 to 20:00 leaves 3 h for breaks, load-out, and return |
| 3 | Cost per package and per stop | $264 ÷ 200; $264 ÷ 160 | $1.32 per package; $1.65 per stop |
| 4 | Density | $264 ÷ 180; ÷ 190; ÷ 220 | $1.47 at 180 packages; $1.39 at 190; $1.20 at 220. Ten packages a route is about 7 cents on every package. |
| 5 | The window check on a dense route | 220 ÷ 1.25 = 176 stops ÷ 20 per hour | 8.8 h. Still inside the 11-h window with 2.2 h of breaks and return; at 260 packages it is not. |

| | | | |
|---|---|---|---|
| 6 | Rural density | If stops per hour fall to 10: 80 stops in 8 h × 1.25 = 100 packages | $2.64 per package. This is where a carrier at $4.50 door-to-door or a local Hub Delivery partner competes with the van, and why the rural network is its own design problem (Part 8). |
| 7 | Attempt versus delivery | The output metric counts an attempt on or before the promised date; a failed attempt re-costs a stop | At 1.5% failed attempts: 90 packages a day at DS-7, 90 × $1.65 = $149 a day of re-attempt, and 90 customers who did not get the unit. |
| 8 | Afternoon Flex wave | Blocks of three to six hours, drivers in their own vehicles, paid per block | About $1.90 per package on a thin afternoon wave; the recovery path for rolled and same-day volume, 44% dearer than the morning route |

## 3.6 The critical pull time chain end to end, and the slack budget

The chain runs backward from the delivery window. Each leg has a planned duration and a tail; each handoff has a buffer; the sum of the buffers is the slack budget, and the buffers are not fungible: slack at the FC dock cannot be spent by a late linehaul. The 2016 MIT thesis defines CPT as the time an item must be picked, palletized, and on a truck (Part 8); the hours here are the reference network.

```
BACKWARD SCHEDULE -- next-day promise, FC-A -> SC-1 -> DS-7, delivered Tuesday by 20:00             [ill

 20:00 Tue  last stop of the route inside the promised window
 09:00 Tue  route departs DS-7                                <- wave; buffer 0.2 h sits before it
 06:30 Tue  station inbound cutoff: last trailer arrives    minus 2.5 h  (unload 0.5, sort 1.3, stage 0.5
 04:00 Tue  trailer departs SC-1                            minus 2.5 h  (second linehaul, 120 miles)
 01:00 Tue  sort-center window opens; FC trailer due        minus 3.0 h  (unload 2.0, sort 0.4, buffer 0.
 21:00 Mon  CPT: trailer leaves FC-A                        minus 4.0 h  (first linehaul, 200 miles, mean
 19:00 Mon  pack complete                                   minus 2.0 h  (sort, stage, load, seal: the do
 17:00 Mon  customer cutoff for next-day on this path       minus 2.0 h  (order-to-pack cycle at p90)
```

**The slack budget, leg by leg**

| LEG | CUTOFF IT PROTECTS | PLANNED | P90 | BUFFER BUILT IN | WHO OWNS THE CLOCK | IF THE BUFFER IS SPENT |
|---|---|---|---|---|---|---|
| Order to pack complete | 19:00 Mon | 1.2 h mean | 2.0 h | 0.8 h inside the 2.0-h cycle allowance | Fulfillment center | The package misses the 21:00 trailer and takes 03:00 Tue, which still makes a Wednesday promise but not Tuesday |
| Pack complete to trailer close | 21:00 Mon | 1.2 h (sort, stage, load) | 1.6 h | 0.4 h of the 2.0-h dock buffer | Fulfillment center outbound | CPT departs with the pile left behind, or departs late and spends the sort center's slack |
| First linehaul | 01:36 Tue latest useful arrival | 4.0 h | 4.45 h | 0.6 h (window 3.0 h against 2.4 h of work) | Transportation; the lane | Trailer rolls to the 07:00 window; its packages take the 10:00 station departure and the afternoon Flex wave at +$0.58 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Sort-center window | 04:00 Tue | 2.4 h for the last trailer | 3.0 h with a door queue | Included above | Sort center | Containers close without the late trailer's packages |
| Second linehaul | 06:30 Tue | 2.5 h | 2.8 h | 0.2 h at the station | Transportation; the lane | Arrival after 06:30: 1,600 packages miss the 09:00 wave; Flex at 14:30 or tomorrow |
| Station sort and stage | 09:00 Tue | 2.3 h for the last trailer | 2.5 h | 0.2 h | Delivery station | Routes leave short; the stragglers go to Flex |
| Route | 20:00 Tue | 8.0 h | 9.5 h | 3.0 h inside the 11-h window | Delivery partner; the catchment | Last stops attempted after the window: a miss on the output metric even though the package was in the van |
| **Total slack on the chain: about 5.4 hours, held in six different places by five different owners.** A configuration change that moves one cutoff moves every upstream cutoff by the same amount. A transit change on one lane does the same thing. That is why a pull-time change and a transit change are the same kind of event and are reviewed on the same page. | | | | | | |

## 3.7 Exceptions as stocks: missort, late, rolled, damaged, and what recovery costs

Each exception is a package that left the main chain and entered a side stock. Each side stock has a recovery path with a cost and a probability of still making the promise. The table prices the reference network's exceptions; the last table assembles them into the output metric.

| EXCEPTION | WHERE (STOCK) | WHAT HAPPENS | RECOVERY PATH | COST PER INCIDENT | PROMISE EFFECT |
|---|---|---|---|---|---|
| Missed CPT | S8, the door | The package is still in the building when its trailer closes. | Next departure on the lane (3 to 6 h later); a recovery truck or air if the promise is worth it. | $0 if the next departure still makes the promise; $2 to $5 if it needs a concession or a recovery truck | Loses the day when the next departure misses the station cutoff. In the reference week, 40% of CPT misses lose the day. |
| Late linehaul | S10, S13 | The trailer arrives after the window or the station cutoff. | Next window; afternoon Flex wave at the station. | +$0.58 per package on the Flex wave; a day late if there is no afternoon wave | Half of late-window trailers lose the day in the reference week. |
| Missort | S12, S14 | The package rides to the wrong station or route. | Re-sort, second linehaul to the right station, next wave. | About $2.00 | 70% lose the day. |
| Rolled at the station | S14 | More packages than routes, vans, or drivers. | Flex wave; extra DSP routes if available; tomorrow's wave. | +$0.58 on Flex; $264 per extra route; a day | Counted at 0.6% of packages in the reference week, all losing the day. |
| Damaged | Anywhere | Pulled from the path; the customer is re-promised or refunded. | A replacement order from the start of the chain. | The replacement's full cost, about $3.31, plus the refund exposure | Speed is affected only through the replacement's own promise. |

| | | | | | |
|---|---|---|---|---|---|
| Failed attempt | S15 → S14 | No access, no safe place, wrong address; the package returns to the station. | Re-attempt next day; return to sender after the attempt limit. | $1.65 per extra stop | Counts as an attempt on time; the customer still waits. Track it separately from the output metric. |
| Carrier late | Carrier path | The carrier misses its own standard on the handed-off package. | None inside the network; the carrier's claim process. | $0 direct; the concession | 3% of carrier packages in the reference week. |

**Assembling the output metric for week 45 of the reference network, 756,000 packages**

| STEP | CAUSE | SHARE × PROBABILITY OF LOSING THE DAY | PACKAGES MISSING THE DAY |
|---|---|---|---|
| 1 | Missed CPT, not recovered | 0.8% × 40% = 0.32% | 2,419 |
| 2 | Late linehaul past a window or cutoff | 1.2% × 50% = 0.60% | 4,536 |
| 3 | Missort | 0.45% × 70% = 0.32% | 2,381 |
| 4 | Rolled at the station | 0.60% × 100% = 0.60% | 4,536 |
| 5 | Carrier late on the 17% handed off | 17% × 3% = 0.51% | 3,856 |
| 6 | Total missing the promised day | 0.32 + 0.60 + 0.32 + 0.60 + 0.51 | 2.34%; 17,728 packages |
| 7 | Delivery estimate accuracy for the week | 1 − 0.0234 | 97.7% attempted on or before the promised date. Illustrative; Amazon has not published a rate. The 2024 MIT thesis names the metric and two of its causes (Part 8). |

**SAY THIS — 120 WORDS**

From the dock the package rides two trailers, one sort, one station sort, and one route. The first linehaul is a distribution, so I plan the pull time on the ninetieth percentile and measure on-time arrival against the window, not the schedule. The sort center is a queue at the doors inside a three-hour window; the marginal trailer is what misses, and a three-person crew is usually cheaper than a door. The station is a wave; what does not sort by nine goes on Flex at a premium or waits a day. Every exception is a side stock with a recovery price and a probability of losing the day, and the output metric is one minus the sum of those.

# Part 4 — One package and one day, walked end to end

Parts 2 and 3 gave the nodes. This part runs them twice: once for a single package, with a timestamp at every stock boundary, and once for the whole 100,000-package day, stage by stage, so that the day's cost and the day's promise both land on one page. Everything here is the reference network ILLUSTRATIVE; the node shapes are public PUBLIC AMAZON PRACTICE.

## 4.1 One package, click to door: timestamps, stocks, and the slack it spent

A two-unit order placed Monday at 14:12 from a customer in DS-7's catchment, promised for Tuesday by 20:00. The path is FC-A → SC-1 → DS-7, the 21:00 CPT. Times are the reference network's; the order is early for its cutoff, so it shows what a normal day looks like, not a rescue.

| CLOCK | STOCK | EVENT | DWELL IN PRIOR STOCK | COST ACCRUED SO FAR | SLACK TO THE NEXT CUTOFF |
|---|---|---|---|---|---|
| Mon 14:12 | S1 | Order placed; promise built from FC-A's 17:00 cutoff + 21:00 CPT + the SC-1 → DS-7 chain. Promise: Tue by 20:00. | — | $0 | 2 h 48 min to the 17:00 customer cutoff (not spent; the order is already in) |
| Mon 14:12 | S1 → S2 | Assigned to FC-A, lane SC-1, CPT 21:00: cheapest feasible path at $2.56 dock-to-door; FC-B would have been $4.06. | < 1 min | $0 | 4 h 48 min to pack complete (19:00) |
| Mon 14:20 | S2 → S3 | Released to pick in CPT priority; batched with 11 other orders into one pick tour. | 8 min | $0 | 4 h 40 min |
| Mon 14:48 | S3 → S4 | Both units picked into a tote (the tour took 28 minutes for 20 units, 43 units per hour on this tour; the building average is 160 per hour, this tour was a long one). | 28 min | $0 | 4 h 12 min |
| Mon 15:10 | S4 → S5 → S6 | Tote conveyed to consolidation; order complete; packed in a box with dunnage, 3 minutes at the station. | 22 min | $0 | 3 h 50 min |
| Mon 15:14 | S6 → S7 | SLAM: weighed (1.9 lb expected, 1.9 lb scanned), labeled, manifested. **The package now exists.** It is committed to lane FC-A → SC-1, CPT 21:00. | 4 min | $0.75 (FC handling, allocated at the label) | Pack complete 3 h 46 min ahead of the 19:00 line: the package is early |
| Mon 15:18 | S7 → S8 | Ship sorter reads the label; chute for SC-1 / 21:00; drops into a gaylord at door 14. | 4 min | $0.75 | 5 h 42 min to CPT |
| Mon 19:40 | S8 | Gaylord closed and loaded into trailer 14-B, position 38 of 2,000. The package has waited at the door 4 h 22 min: the longest stock on its whole path inside a building. | 4 h 22 min | $0.75 | 1 h 20 min to CPT |
| Mon 21:05 | S8 → S9 → S10 | Trailer closes at 21:00 with 1,590 packages (79.5% fill), seals, departs 21:05. Five minutes of the 36-minute arrival slack are spent at the gate. | 1 h 25 min | $1.15 (+$0.40 linehaul: $640 ÷ 1,590 = $0.403) | 31 min of arrival slack remain |

| | | | | | |
|---|---|---|---|---|---|
| Tue 01:17 | S10 → S11 | Arrives SC-1 after 4 h 12 min on the road (the lane's p70). Door 9 is free; unload starts 01:20. | 4 h 12 min | $1.15 | 19 min of arrival slack remain; 2 h 40 min to the 04:00 window close |
| Tue 02:05 | S11 → S12 | Its gaylord comes off the trailer; inducted 02:10; sorter sends it to the DS-7 chute 02:11; containerized into a DS-7 bag 02:40; bag loaded to the DS-7 trailer 03:30. | 48 min at the door; 1 h 25 min in sort WIP | $1.45 (+$0.30 sort-center touch) | 30 min to the window close |
| Tue 04:00 | S12 → S13 | SC-1 → DS-7 trailer departs with 1,600 packages. | 30 min staged | $1.69 (+$0.24) | 2 h 30 min to the 06:30 station cutoff; transit mean 2.5 h; the plan carries the 0.2-h buffer at the station, not here |
| Tue 06:25 | S13 → S14 | Arrives DS-7 five minutes early. Unloaded 06:50; inducted 07:05; sorted to route 17, stop 92 of 160; staged in the route 17 bag 08:30. | 2 h 25 min | $1.99 (+$0.30 station handling) | 30 min to the 09:00 wave after staging |
| Tue 09:04 | S14 → S15 | Driver checks in 08:40, loads out 08:50, departs 09:04 with 198 packages on 158 stops. | 34 min | $3.33 (+$1.33: $264 ÷ 198) | 10 h 56 min to the 20:00 window |
| Tue 13:53 | S15 → S16 | Stop 92 reached at 13:52 (91 stops ÷ 20 per hour = 4 h 33 min plus a 15-min break); delivered to the porch, photo taken. Attempt on the promised date: counted. | 4 h 49 min on the van | $3.33 total; $2.58 dock-to-door | Delivered 6 h 07 min inside the window |

**What the package's day adds up to**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Click to door | Mon 14:12 → Tue 13:53 | 23 h 41 min |
| 2 | Time being worked on | Pick 2 min of the tour + pack 3 + SLAM 0.2 + sort 0.1 + unload share 2 + induct and sort 1 + containerize 1 + station sort 1 + load-out 2 + the stop 1 | About 13 minutes. Under 1% of the elapsed time. |
| 3 | Time in transit (pipeline stocks S10, S13, S15) | 4 h 12 min + 2 h 25 min + 4 h 49 min | 11 h 26 min; 48% of elapsed |
| 4 | Time waiting for a cutoff (S8 at the FC door, S12 staged, S14 staged) | 5 h 47 min + 30 min + 34 min | 6 h 51 min; 29% of elapsed. This is the price of a scheduled network: packages wait for departures. It is not waste; it is what makes the trailer full. |
| 5 | Time in work queues (S2, S3, S4, S5, S11, sort WIP, station WIP) | The remainder, which includes the 13 minutes of work | About 5 h 24 min; 23% |
| 6 | Slack spent against the chain's 5.4 hours | 5 min at the gate + 12 min of transit over mean + 0 at the station | About 17 minutes. A quiet Monday. On the plus-sixty day in 2.5 the same package would have spent all of it before it reached the dock. |

| | | | |
|---|---|---|---|
| 7 | Cost by owner | FC $0.75; linehaul 1 $0.40; sort center $0.30; linehaul 2 $0.24; station $0.30; route $1.33 | $3.33; dock-to-door $2.58. The last mile (station plus route) is $1.63, 63% of dock-to-door. The two sorts together are $0.60, less than half the route. |

**SAY THIS — 111 WORDS**

A package is worked on for about thirteen minutes and travels for about eleven hours; the rest of its day it waits for a cutoff or sits in a queue. The waiting for cutoffs is what fills the trailer, so I do not try to remove it; I try to make sure it is the right amount at the right door. The queue time is what I attack, because that is where the slack goes on a bad day. And the money is at the end of the chain: the route alone is half of dock-to-door cost, which is why density and the station catchment matter more than any single sort.

## 4.2 The 100,000-package day: eight stages, every number, one-page close

The same day, seen from above. Each stage lists what flowed, what it cost, the capacity it used, and the cutoffs it had to make. The paths are the four from 2.1: 44,800 packages via SC-1, 25,600 via SC-2, 15,000 direct to five stations, 14,600 handed to carriers at the dock.

| # | STAGE | WHAT FLOWED | CAPACITY USED | COST | CUTOFFS MET |
|---|---|---|---|---|---|
| 1 | Orders and promises | 100,000 packages' worth of orders over 24 hours, peaking 11:00–14:00; 4 CPTs (03:00, 09:00, 15:00, 21:00) of about 25,000 each plus later-CPT fill work | Software; the promise table | — | Customer cutoffs by path: 17:00 for the SC-1 next-day path; 18:30 direct; 15:30 SC-2 (longer station chain) |
| 2 | Pick | 160,000 units | 1,000 direct hours; 1,150 paid; 58 pickers a shift; 160 units per hour | $25,300 ($0.25 per package) | Pick backlog never above 1.2 hours; urgent work cleared each CPT by 14:06, 08:06, 02:06, 20:06 |
| 3 | Pack and SLAM | 100,000 packages; 1,700 weight mismatches (1.7%) held for check, 1,520 cleared, 180 re-picked | 1,000 direct pack hours; 58 packers a shift; SLAM 3 lines, 93% at the 13:00 peak | $25,300 labor + $15,000 materials ($0.40) | Pack complete 1.9 h ahead of each CPT on the base day |
| 4 | Ship sort and dock | 100,000 sorted to 11 lanes × 4 CPTs plus 6 carrier pickups; 54 own trailers loaded at 1,581 average (79.1% of positions); 53 departed on time, 1 departed 00:20 late with 1,400 packages | Sorter 50% average, 70% peak; 200 load hours; 19 live doors of 24 | $9,460 ($0.09) | CPT compliance 53 ÷ 54 = 98.1% |
| 5 | First linehaul | 28 trailers to SC-1 (44,800), 16 to SC-2 (25,600), 10 direct (15,000); 6 carrier pickups (14,600) | 15 tractors on the SC-1 lane alone; 54 trailer-departures | 28 × $640 + 16 × $480 + 10 × $288 = $28,480 ($0.33 blended on the 85,400 own-network packages) | On-time arrival 52 ÷ 54; one SC-1 trailer arrived 01:52 (p99 transit) and rolled to 07:00; one late departure rolled |

| | | | | | |
|---|---|---|---|---|---|
| 6 | Sort centers | SC-1: 100,000 in four windows (44,800 from FC-A, the rest from two other FCs); SC-2: 60,000 (25,600 from FC-A) | SC-1: 12 doors at 87%, 10 inducts, sorter 42%, 786 paid hours; SC-2: 8 doors, 6 inducts | SC-1 $0.30 × 44,800 = $13,440; SC-2 $0.31 × 25,600 = $7,936 | SC-1 missort 0.3% = 300 across all origins; 2 trailers' worth (3,200 packages) rolled a window, both from the one late arrival and one door queue |
| 7 | Second linehaul and stations | SC-1 → 20 stations averaging 5,000; SC-2 → 12 stations; DS-7 received 6,000 on 4 trailers (3 from SC-1, 1 from SC-2) | About 60 station trailers from SC-1 a day at 1,600 planned | $0.24 × 44,800 + $0.24 × 25,600 = $16,896; station handling $0.30 × 85,400 = $25,620 | Station cutoffs made on 58 ÷ 60 SC-1 trailers; the two late ones put 3,200 packages on the afternoon waves |
| 8 | Routes and doors | 85,400 own-network packages on 427 routes of 200; 14,600 carrier; 0.6% (512) rolled at stations for want of routes; 1.5% failed first attempt | 427 vans and drivers across 37 stations; afternoon Flex blocks for 3,712 packages | 427 × $264 = $112,728 + Flex premium 3,712 × $0.58 = $2,153 + carrier 14,600 × $4.50 = $65,700 | 97.7% attempted on or before the promised date |

**The one-page close for the day**

| LINE | ACTUAL | PLAN | NOTE |
|---|---|---|---|
| Packages shipped | 100,000 | 100,000 | Reference day; the plan was the forecast at T-1 |
| Fulfillment-center handling | $75,060 — $0.75 per package | $0.75 | Labor $0.60, materials $0.15 |
| Own-network dock-to-door | $207,253 on 85,400 packages — $2.427 | $2.402 | Linehaul 1 $28,480; sorts $21,376; linehaul 2 $16,896; stations $25,620; routes $112,728; Flex premium $2,153 (the whole overrun) |
| Carrier handoff | $65,700 on 14,600 packages — $4.50 | $4.50 | Rate card; 14.6% of volume, 24% of dock-to-door dollars |
| **Transportation cost per package, dock to door** | **$272,953 ÷ 100,000 = $2.730** | **$2.708** | +$0.022, +0.8%; all of it is the Flex premium on the 3,712 packages that rolled a window or a wave. Fill and route density were at plan. |
| All-in outbound per package | $3.480 | $3.458 | Handling plus dock-to-door |
| CPT compliance | 53 ÷ 54 = 98.1% | ≥ 98% | One trailer 20 minutes late: a pack backlog at the 21:00 CPT |
| Trailer fill at close | 79.1% | 79.1% | Positions: 80% on sort-center lanes, 75% direct; cube about 32%; weight about 9% |
| Linehaul on-time arrival | 52 ÷ 54 = 96.3% | ≥ 97% | One p99 transit; one late departure |
| Sort-center missort | 0.30% | ≤ 0.30% | 300 packages across SC-1's day |
| Rolled at stations | 0.6% — 512 packages | ≤ 0.5% | Two stations short of routes on a Monday |

| | | | |
|---|---|---|---|
| Packages per route | 200 | 200 | 427 routes |
| **Delivery estimate accuracy** | **97.7%** | **≥ 97.5%** | Causes in 3.7: late linehaul 0.60, rolled 0.60, carrier 0.51, missed CPT 0.32, missort 0.32 points |

**SAY THIS — 123 WORDS**

On a hundred-thousand-package day the building spends about seventy-five cents a package getting the package to the dock and the network spends two dollars seventy getting it to the door. On the packages we carry ourselves, dock-to-door splits roughly a fifth linehaul, a tenth sorts, a tenth stations, and more than half routes; the carrier handoff is fifteen percent of the packages but a quarter of the dollars. The day's promise rate is one minus the exceptions: late trailers, rolled packages, carrier misses, missed pull times, missorts, in that order. When I read the close I look at two things first: fill against plan, because it is the fastest cost lever, and rolled volume, because it is tomorrow's promise miss already visible today.

# Part 5 — Doing the job: the eight responsibilities, step by step

Each section takes one line of the job description and answers the question the interviewer is actually asking: what would you do on Monday, with what inputs, what model, what decision rule, and what proof. Every section has the same six pieces: the deliverable, the steps, the worked math on the reference network, where the decision gets made, the trap, and the smaller version of the same work that is on the record. Record items carry their status from the numbers card (8.3) RECORD USER ANCHOR; the worked math is the reference network ILLUSTRATIVE.

## 5.1 Design and roll out a new outbound network model, and prove cost and speed

**The deliverable.** A network model is a rule for which packages take which path: through a hub, a last-mile sort center, directly from the fulfillment center to the station, from a same-day site, or to a carrier. "New" means a different rule for some set of origin-destination pairs. The deliverable is the rule, the stations and lanes it applies to, the cost and speed it produces against the current rule on the same packages, and a rollout plan with gates. Amazon's public version of this is regionalization and the direct fulfillment-center-to-station trucks that became easier to fill under it PUBLIC AMAZON PRACTICE, Part 8.

| # | STEP | WHAT YOU DO | OUTPUT |
|---|---|---|---|
| 1 | Baseline the current path at the package level | For every origin-destination pair, the current path, its cost by leg, its chain hours, its on-time rate, over the last 13 weeks. Reconcile the sum to the ledger before anything else. | A path table: 11 lanes × 37 stations for FC-A, with TCPP and hours per pair |
| 2 | Write the candidate rule | Direct inject: any station receiving at least N packages a day from FC-A gets a direct trailer at one or two departures instead of riding through SC-1. N comes from the breakeven in the math below. | A rule with one parameter, and the list of stations it selects |
| 3 | Model both rules on the same demand | Run the 13 weeks of actual packages through both rules. Hold labor rates, trailer costs, and route density fixed so only the path changes. Count trailers, doors, tractors, sort-center volume lost, and the station's inbound profile. | Cost delta per package per station; chain hours delta; capacity consumed and freed |
| 4 | Price the second-order effects | SC-1 loses volume, so its fixed cost per package rises on what remains. FC-A needs more live doors. The direct trailer's fill is thinner than the hub trailer's. Put each in the model with its own line. | A net number, not a gross one |
| 5 | Set the gates and the control | Pick one station as the pilot and three similar stations as controls. Define the gates: fill at or above plan for three weeks, on-time arrival at or above the control, TCPP for the pair at or below the modeled number within 5%. | A pilot charter one page long |
| 6 | Roll out in rings | One station for four weeks; then five; then the rule. Each ring re-runs the model with the measured fill and transit, not the assumed ones. Stop if a gate fails twice. | A calendar with the configuration changes each ring needs (Part 5.6) |
| 7 | Prove it | Difference in differences: (pilot after − pilot before) − (control after − control before), on TCPP and on-time, for the same weeks. Report the measured number next to the modeled one. | The proof page, with the model's error stated |

> **Direct inject from FC-A: the breakeven and the proof on the reference network**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Middle mile via SC-1, per package | Linehaul 1 $0.40 + sort $0.30 + linehaul 2 $0.24 | $0.94 (station handling and the route are the same on both paths) |
| 2 | Direct trailer, 90 miles | $288 per departure ÷ packages on it | $0.19 at 1,500; $0.29 at 1,000; $0.38 at 750; $0.58 at 500 |
| 3 | Breakeven packages per departure | $288 ÷ $0.94 | 306 packages, 15% of positions. Any station receiving more than about 300 a day from FC-A per planned departure is cheaper direct. |
| 4 | Cost advantage at plan fill | $2.56 − $1.81 | $0.75 per package, 29% of dock-to-door on the pair; on 15,000 direct packages a day, $11,250 a day, about $4 million a year |
| 5 | Speed advantage | Chain hours via SC-1: 4.0 + 3.0 + 2.5 = 9.5 h from CPT to station cutoff. Direct: 2.0 h. | 7.5 hours. Either the customer cutoff on the pair moves from 17:00 to about 00:30 with the same promise, or the same cutoff reaches a one-day-faster promise where the station has an earlier wave. |
| 6 | Second-order: SC-1 fixed cost on the remaining volume | $13,000 ÷ 85,000 − $13,000 ÷ 100,000 (if 15,000 leaves) | +$0.023 on 85,000 packages = $1,950 a day against the $11,250 |
| 7 | Second-order: FC-A doors and yard | 10 direct departures × 1.5 h loading ÷ 24 h, at 85% | Under one additional live door on average, but 5 concurrent at each direct CPT; the 24-door building has 5 spare (2.4) |
| 8 | Net | $11,250 − $1,950 | $9,300 a day, $3.3 million a year, and 7.5 hours on 15% of the building's packages. The proof page reports the measured version of this line. |
| 9 | Minimum detectable effect in the pilot | Pair TCPP standard deviation week to week about $0.08; 4 weeks pilot, 3 controls | A $0.75 effect against $0.08 noise is detectable in two weeks; the four weeks are for fill and on-time to settle, not for statistics |

**Where it gets decided.** The rule and the ring plan go to the network design review with transportation finance and the regional operations director in the room; the configuration change itself is a scheduled event on the change calendar with the fulfillment center, the sort center, and the station each signing the one-page in 5.6. **The trap.** Proving the gross saving and not the net: the sort center's lost volume, the thinner direct trailer, the extra live door, and the station's earlier inbound all cost something, and a model that leaves them out gets found out by finance in the second meeting rather than the first.

**The small version I have run.** The Elementum lane program `RECORD`: measured 200 lanes at the shipment level, found that lead-time variance was most of the buffer on the worked lane, re-set the buffers from the measured distribution, and released about 580,000 units, about $1.5 million one time, at 99 percent service. Same shape as step 1 and step 3: baseline at the unit of work, change one rule, hold everything else, measure on the same lanes. The scale here is a lane table of two hundred rows; the job's scale is thousands of origin-destination pairs. The method does not change; the tooling does, and that is what the data-science partnership in 5.6 is for.

## 5.2 Profitability of sortation capacity by region and business line; facility design standards

**The deliverable.** For every sort center, a statement of whether it earns its cost against the best alternative use of its packages, by region and by business line (standard parcel, small and flat, large and non-conveyable), and from the pattern across sites, the design standards for the next building: sorter rate, doors, non-conveyable handling, and the volume band where the type works.

| # | STEP | WHAT YOU DO | OUTPUT |
|---|---|---|---|
| 1 | Define "earns its cost" | Contribution = (cost of the best alternative path for the same packages − variable cost through this site) × packages, minus the site's fixed cost. The alternative is not "nothing"; it is the carrier, the neighboring sort center, or a direct lane. | One formula, two alternatives per site |
| 2 | Build the variable and fixed cost per site from the ledger | Labor hours × wage by function; materials; lease, depreciation, utilities, salaried staff as fixed. Reconcile to the site's monthly P&L within 2%. | A cost card per site |
| 3 | Price the alternatives per package | Carrier rate card by zone; the neighbor's variable cost plus the longer linehauls; the direct lane at its fill. | Alternative cost per package by destination |
| 4 | Compute breakeven volume and the cushion | Breakeven = fixed cost ÷ contribution per package. Cushion = actual volume ÷ breakeven. | A ranked list of sites by cushion |
| 5 | Split by business line | Allocate sorter time, trailer positions, and labor by the line's cube and conveyability. A bulky package uses 6 to 8 standard positions and cannot ride the sorter. | Middle-mile cost per package by line; which lines the site should not touch |
| 6 | Add the speed term | Packages that would lose a day on the alternative path, priced at the concession-and-demand value finance agrees to, or shown as a count alongside the dollars. | Dollars and a day count, kept separate |
| 7 | Turn the pattern into standards | Across sites: at what volume does a last-mile sort center beat direct lanes; what sorter rate for what peak window; how many doors per trailer per window; a non-conveyable bypass; modular build. | A design-standards page with the math behind each line |

**Does SC-2 earn its cost? 60,000 packages a day, fixed cost $8,400 a day**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Variable cost through SC-2, dock to door | Linehaul 1 $0.30 + sort labor $0.17 + linehaul 2 $0.24 + station $0.30 + route $1.32 | $2.33 per package |
| 2 | Alternative A: hand the packages to a carrier | $4.50 − $2.33 | Contribution $2.17; breakeven $8,400 ÷ $2.17 = 3,871 a day; cushion 60,000 ÷ 3,871 = 15.5×. Against the carrier the building is not in question. |
| 3 | Alternative B: route the packages through SC-1 | SC-1's variable on these packages is $2.59 (longer first linehaul $0.56 for the extra miles, $0.17 sort, $0.24, $0.30, $1.32); $2.59 − $2.33 | Contribution $0.26; breakeven $8,400 ÷ $0.26 = 32,308 a day; cushion 1.86×. Net $15,600 − $8,400 = $7,200 a day, about $2.6 million a year. |

| | | | |
|---|---|---|---|
| 4 | Speed term for alternative B | SC-1's chain is 2 hours longer for SC-2's stations; 20% of SC-2's packages would miss their current promise | 12,000 packages a day lose a day. This goes on the page as a count. Finance may price it; the model does not invent the price. |
| 5 | Verdict | Cushion 1.86× against the real alternative, plus 12,000 a day of speed | Earns its cost. Watch item: if volume falls below about 40,000 the cushion is 1.2× and the question reopens. |

**By business line through SC-1: middle-mile cost per package**

| LINE | POSITIONS PER TRAILER | LINEHAUL 1 | SORT | LINEHAUL 2 | MIDDLE MILE | READING |
|---|---|---|---|---|---|---|
| Standard parcel | 1,600 at plan | $0.40 | $0.30 | $0.24 | $0.94 | The design case |
| Small and flat, bagged | 6,400 (bags of 25 in a position) | $0.10 | $0.22 (bag induct; fewer touches) | $0.06 | $0.38 | Sort them in bags to the station and let the station open the bag: the standard that follows is a bag-sort induct line |
| Large, non-conveyable | 240 (one in 6.7 positions) | $2.67 | $0.95 (manual sort, two-person handling) | $1.60 | $5.22 | Versus a dedicated bulky lane direct to a bulky station: $288 ÷ (300 × 80%) + $0.60 handling = $1.80. The sorter building loses $3.42 on every bulky package it touches. |

**The design standards that fall out:** (1) size the sorter at the peak window's standard-parcel rate ÷ 0.85, not the daily average: SC-1 needs 8,333 ÷ 0.85 = 9,800 per hour at today's volume, so its 20,000 sorter is right for a 2× growth plan and wrong for a flat one; (2) doors = trailers in the peak window × unload hours ÷ window hours ÷ 0.85: 16 × 2 ÷ 3 ÷ 0.85 = 12.5, so SC-1's 12 doors are at the edge and the next building gets 14 for the same volume; (3) a non-conveyable bypass dock, or none of that line in the building; (4) a bag-sort induct line when small and flat exceeds about a quarter of volume; (5) build in two phases on one yard so the second sorter is a permit, not a site search.

**Where it gets decided.** The site verdicts go to the annual network capacity review with finance and the regional directors; the design standards go to the facility design authority as a versioned document with the math attached. **The trap.** Comparing the site to "nothing" instead of to the alternative path. Every sort center beats a carrier; the question is whether it beats its neighbor and the direct lane, and that cushion is usually under 2×.

**The small version I have run.** The Flexport cost-per-container work RECORD ENDPOINTS: $0.66 to $0.03 a container, with the worksheet rates behind it labeled illustrative and the finance rate for one interaction ($0.18) kept separate from the ticket price. That is the step-2 discipline: build the variable cost from the ledger's own rates, label what is assumed, reconcile to the endpoint finance already believes. The Elementum spend cut USER ANCHOR, 597 lanes as 80 percent of spend, is step 4's ranking logic: do the sites in cushion order, not alphabetical order.

## 5.3 Capital plans and business cases for new and expanded sites, defended with finance

**The deliverable.** A business case is a cash-flow model with a sizing behind it and a decision page in front of it: what we build, how big, where, what it earns against the current network, what has to be true for the number to hold, and when we would stop. Defending it with finance means every line can be traced to a rate, a count, or a labeled assumption, and the sensitivities are on the page before finance asks for them.

| # | STEP | WHAT YOU DO | OUTPUT |
|---|---|---|---|
| 1 | State the problem in packages and hours | Which packages are expensive or slow today, and why: of the north metro's 60,000 a day, 40,000 ride 250 miles to SC-1 and 180 miles back out, and 20,000 go to a carrier because the chain is too long for the promise. | The demand the site would serve, by path, today |
| 2 | Size for the year-five peak window, not today's average | Volume growth, peak-to-average ratio, the window share, the sorter and door math from 5.2. Phase the build so year-one capital matches year-one volume. | Sorter rate, doors, square feet, yard positions, phase plan |
| 3 | Build the operating case | Saving per package by path × packages, minus the new site's fixed opex, minus cannibalization at the sites that lose volume, plus or minus the speed term as a count. | Net daily and annual saving, with each piece on its own line |
| 4 | Build the capital case | Capex by phase, the ramp curve, ten-year cash flows, the discount rate finance gives you, NPV, IRR, payback, and the three sensitivities that matter: volume, capex, contribution per package. | The cash-flow page |
| 5 | Compare the alternatives honestly | Lease versus build; expand SC-1 instead; renegotiate the carrier; do nothing and accept the cost. Each on the same cash-flow basis. | An alternatives table with the reason each loses |
| 6 | Write the kill criteria | What measured number at month six or month twelve would mean the case was wrong, and what we would do then. | Two triggers, each with an action |
| 7 | Pre-wire finance and operations | Walk the controller through the rates and the ramp before the review. Walk the regional director through the labor plan and the launch calendar. Change the page where they are right. | A case that survives the room because it was already argued |

**SC-3, a new last-mile sort center for the north metro: the case on the reference network**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Today's cost for the 40,000 via SC-1 | Linehaul 1 at 250 miles $800 ÷ 1,600 = $0.50; sort $0.30; linehaul 2 at 180 miles $576 ÷ 1,600 = $0.36; station $0.30; route $1.32 | $2.78 per package |
| 2 | Through SC-3 at 100 miles from FC-A, 40 miles to the stations | $0.20 + $0.17 variable sort + $0.24 (short lane, lighter trailers) + $0.30 + $1.32 | $2.23 variable; saving $0.55 on the 40,000 that re-route |
| 3 | The 20,000 carrier packages brought in-network | $4.50 − $2.23 | $2.27 per package saved |
| 4 | Gross daily saving | 40,000 × $0.55 + 20,000 × $2.27 = $22,000 + $45,400 | $67,400 a day |
| 5 | Less SC-3 fixed opex | Lease, depreciation excluded (in capex), utilities, salaried staff: $13,000 a day | $54,400 |
| 6 | Less cannibalization at SC-1 | SC-1 falls to 60,000; fixed per package $13,000 ÷ 60,000 − $0.13 = $0.087 on 60,000 | $5,200 a day; net $49,200 a day; × 360 = $17.7 million a year at run rate |
| 7 | Capex and payback | Building fit-out, sorter, yard, systems: $45 million; $45M ÷ $17.7M | 2.5 years simple payback |
| 8 | NPV and IRR, ten years at 10%, year-one ramp at 50% | Cash flows −45; 8.9; 17.7 × 9 with no growth credit | NPV about $56 million; IRR about 32% |

| | | | |
|---|---|---|---|
| 9 | Sensitivities | Volume −25%: NPV $24 million. Capex +20%: $47 million. Contribution −$0.15 per package: $37 million. All three at once: about $1 million, roughly breakeven. | The case is robust to any one of them and breakeven against all three together: that sentence goes on the decision page |
| 10 | Speed term | 60% of the north metro's 60,000 gain a day on the shorter chain | 36,000 packages a day, stated as a count next to the dollars |
| 11 | Sizing | Year-five volume 60,000 × $1.08^5$ = 88,160; peak day 1.6× = 141,055; peak window 35,264 in 3 hours; sorter 35,264 ÷ 3 ÷ 0.85 = 13,829 per hour | A 15,000-per-hour sorter; 22 trailers per peak window → 22 × 2 ÷ 3 ÷ 0.85 = 17.3, so 18 doors; phase 1 at 10,000 per hour and 12 doors on a yard permitted for 18 |
| 12 | Kill criteria | Month six: in-network share of the north metro below 80% of plan, or measured contribution per package below $0.40 on the re-routed 40,000 | Action: hold phase 2; re-run the case with measured rates; if still below, re-purpose the building as a direct-lane cross-dock |

**Where it gets decided.** Capital committee, with transportation finance owning the cash-flow page and the regional vice president owning the launch risk. Your job in the room is the sizing, the saving by path, and the kill criteria. **The trap.** A case that is right on average and wrong at the peak: the sorter and the doors are sized for the peak window of year five, and if the page shows a daily average, finance will approve a building that rolls packages every December.

**The small version I have run.** The Flexport roadmap re-sequencing RECORD: value in month six instead of month eighteen, 66 percent less effort, about $5 million avoided once; the model under it (twelve engineers, $20,000 a month, 20,000 shipments) is MODELED, not measured, and I say so. The Austria program RECORD, EXECUTED: about $500,000 a year at one site, then about $8 million across 25 sites, with the $5 million base a USER ANCHOR. Both are the step-3 and step-7 muscles: a saving built per unit and multiplied by a count, pre-wired with the people who own the count. The capital-committee format itself, ten-year cash flows and a capex line, is one I have modeled and defended at the program level, not at the building level; I would lean on transportation finance for the discount rate and the depreciation treatment and own the operating case.

## 5.4 Design and run dispatch pilots, then scale what works

**The deliverable.** A dispatch strategy is a rule about when a trailer or a route leaves and what it waits for. A pilot is the rule, run on a defined set of lanes or stations for a defined period, against controls, with a pre-registered success metric, guardrails, a kill switch, and a scale decision at the end. "Scale what works" means the scale decision is made by the number the charter named, not by the enthusiasm in the room.

| # | STEP | WHAT YOU DO | OUTPUT |
|---|---|---|---|
| 1 | Write the hypothesis as a rule with a trigger | "If the marginal trailer at a CPT is under 70% fill and the lane's measured slack is at least 45 minutes, hold up to 30 minutes and fill from the next CPT's ready work." A rule a dock lead can apply without a model. | One sentence, two thresholds |
| 2 | Compute the expected value before you run anything | Probability the trigger fires × (probability it saves a departure × trailer cost − probability it makes the trailer late × the cost of late). | Dollars per lane-day; whether the pilot is worth its cost |
| 3 | Choose the unit, the arms, and the sample | Unit: lane-day. Arms: treated lanes and matched controls, same sort center, same window. Sample: from the fill variance and the effect you need to see. | Lanes per arm and days to run |

| | | | |
|---|---|---|---|
| 4 | Pre-register the metrics | Primary: fill at close. Guardrails: on-time arrival at the sort center, CPT compliance, packages rolled. Kill switch: on-time arrival more than one point below the control for two consecutive days. | The charter page, signed by the sort-center and FC leads |
| 5 | Run it with a daily flash | Every morning: trigger fires, holds taken, fill, on-time, rolls, for each arm. Watch for gaming: a dock that holds every trailer to hit fill. | A one-screen flash and a log of every hold |
| 6 | Read it | Difference in fill between arms with its confidence interval; the guardrails; the dollars at the measured rates. Compare to the step-2 expected value and explain the gap. | The read-out: scale, change, or stop |
| 7 | Scale in rings with the rule in the system | From a dock lead's judgment to a configured hold rule with the same thresholds, so the behavior is the rule and not the person. Ring 2 on 20 lanes; ring 3 the region. | A configured rule and a monitoring line on the scorecard |

**Pilot: dynamic hold on the marginal trailer, FC-A → SC-1**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | How often the trigger fires | Share of CPTs where the last trailer is under 70% and slack is at least 45 minutes, from 13 weeks of history | 35% of lane-days |
| 2 | Value when it works | The hold lets the marginal trailer absorb the next CPT's early work; 60% of the time that removes one departure later in the day | 0.6 × $640 = $384 |
| 3 | Cost when it goes wrong | 5% of holds make the trailer miss the sort window; half its 1,600 packages lose a day at about $2 of recovery and concession | 0.05 × 1,600 × 0.5 × $2 = $80 |
| 4 | Expected value per lane-day | 0.35 × ($384 − $80) | $106; about $38,000 per lane-year; about $770,000 a year on 20 lanes. Worth a pilot; not worth a program until it is measured. |
| 5 | Sample size for a 5-point fill effect | n per arm = $2 \times (1.96 + 0.84)^2 \times \sigma^2 \div \delta^2$ with $\sigma = 0.20$ (fill standard deviation across lane-days) and $\delta = 0.05$ | 251 lane-days per arm: 10 lanes per arm for 25 days. If σ is 0.30, 564 lane-days, so measure σ first. |
| 6 | Guardrail arithmetic | Control on-time arrival 96.3%; kill if treated is below 95.3% two days running | On 10 lanes × about 7 departures = 70 trailers a day, one point is 0.7 trailers: the kill switch needs the two-day rule or it fires on noise |
| 7 | Read-out example | Treated fill 84.1% vs control 79.6%, interval ±1.9 points; on-time 96.0% vs 96.3%; departures saved 0.55 per lane-day | Effect 4.5 points, real; departures saved 0.55 vs the 0.6 × 0.35 = 0.21 assumed: the hold removes more departures than modeled because the next CPT's early work was larger than assumed. Scale, with the model corrected. |

**Where it gets decided.** The charter is approved by the sort-center and fulfillment-center general managers whose docks run it; the scale decision is the weekly network review with the read-out page. **The trap.** No control arm. Fill moves with volume, day of week, and the weather; a before-and-after on the treated lanes alone will show an effect that is half seasonality, and the scaled program will disappoint.

**The small version I have run.** The Utah robotics gate RECORD: a 98 percent bar, week two at 94 percent, about 30 percent of volume held back, full in week three. That is step 4 and step 7: a pre-registered bar, a throttle instead of a switch, and a scale decision made by the number. The Sephora available-for-sale rollout RECORD FOR THE OUTCOME had rollback triggers on the configuration; the Flexport wave hold at 76 percent arrival accuracy against an 80 percent bar RECORD is a pilot that did not pass its gate and was held, which is the half of this responsibility people forget to tell.

## 5.5 Define, track, and report transportation cost per package and performance against plan

**The deliverable.** A metric is a definition, a numerator, a denominator, a time grain, an owner, a source, and a reconciliation rule. Transportation cost per package is the one this job is named for; "performance against plan" is the set of variances that explain why the number moved. The full scorecard is Part 6; this section is the definition work that has to be done first.

**Transportation cost per package: the definition, line by line**

| ELEMENT | RULE | REFERENCE NETWORK |
|---|---|---|
| Numerator | All cost from the fulfillment-center dock to the customer's door: first linehaul (own fleet at cost, contracted at invoice), sort-center operating cost (labor, lease, utilities, depreciation), second linehaul, delivery-station operating cost, last-mile route cost (DSP invoices, Flex block payments), carrier invoices for handed-off packages, and recovery cost on exceptions (re-sorts, recovery trucks). Excludes fulfillment-center handling, inbound, returns transportation, and concessions, each of which is its own line. | $272,953 on the reference day |
| Denominator | Packages with a delivery attempt in the week, by the station that attempted them. Not packages shipped (timing mismatch), not units (mix noise), not orders (multi-package orders). | 100,000 |
| Grain | Week, by region, by path (hub, last-mile sort center, direct, same-day, carrier), by business line. Daily flash on the operational inputs only; dollars weekly. | W45: $2.834 blended; $2.665 SC-1 path; $4.50 carrier |
| Accrual rule | Carrier and DSP invoices lag two to four weeks. Accrue at the rate card × measured volume in the week; true up when the invoice posts; show the true-up as its own line so it does not look like a rate change. | Reference W45 carries $14,000 of true-up from W42 |
| Allocation rule | Shared trailers: by packages on board. Sort-center fixed cost: by packages sorted in the week (so a volume drop shows as a rate rise at that site, which is the truth). Stations: by packages attempted. | SC-1 fixed $0.13 at 100,000; $0.22 at 60,000 |
| Plan | Volume plan by path × rate plan by path, set at the annual plan and re-forecast quarterly. The plan is a number per path, not one blended number, or mix will hide inside rate. | Plan $2.708 blended; $2.56 SC-1 path |
| Variance decomposition | Actual − plan = volume variance (plan rate × volume change, in dollars) + rate variance (volume × rate change, by path) + mix variance (path share change × plan rate). Worked in 6.4. | W45 vs W44: mix +$0.050, rate +$0.070 |

| | | |
|---|---|---|
| Parity | Before the first scorecard: the metric's weekly dollar total must tie to the ledger within 1%, and the package count to the delivery system within 0.5%. Until it does, the scorecard is not published. | The gate the Flexport review used (below) |

**Network performance against plan: the companion metrics and their definitions**

| METRIC | NUMERATOR ÷ DENOMINATOR | GRAIN AND OWNER | PLAN ON THE REFERENCE NETWORK |
|---|---|---|---|
| Delivery estimate accuracy | Packages with an attempt on or before the promised date ÷ packages promised (definition from the 2024 MIT thesis, Part 8) | Daily flash; weekly; the region | ≥ 97.5% |
| CPT compliance | Trailers departed within 15 minutes of the pull time ÷ trailers scheduled | Daily; the fulfillment center | ≥ 98% |
| Trailer fill at close | Positions used ÷ positions available, with cube and weight shown beside it | Daily by lane; transportation | 80% sort-center lanes, 75% direct |
| Linehaul on-time arrival | Trailers arrived before (window close − unload − sort) ÷ trailers due | Daily by lane; transportation | ≥ 97% |
| Sort window completion | Packages containerized before the window close ÷ packages received for the window | Per window; the sort center | ≥ 99% |
| Missort rate | Packages arriving at the wrong station or route ÷ packages sorted | Daily; the sort center | ≤ 0.30% |
| Rolled volume | Packages received for a wave not dispatched in it ÷ packages received | Daily; the station | ≤ 0.5% |
| Packages per route | Packages dispatched ÷ routes dispatched | Daily; the station and the DSP | 200 |
| Volume vs plan | Packages attempted ÷ planned packages, by path | Weekly; planning | ±3% is noise; beyond it the plan is wrong, not the network |

**Where it gets decided.** The definitions are signed by transportation finance (dollars), the operations directors (counts), and the data owner (sources) in one session, and frozen; a change to a definition is a dated version, never a silent edit. **The trap.** Publishing a cost per package whose denominator moves with the source: shipped one week, attempted the next. The number will look volatile and nobody will trust the trend.

**The small version I have run.** The Flexport weekly review `RECORD`: seven metric definitions agreed and frozen before the first page was published; system accuracy measured 60 to 90 percent with parity within one percent against the source; review time from 30 to 6 hours a week, priced at $103,680 at an assumed $90 an hour `RECORD-FROM-DOCS ARITHMETIC, PRICED NOT SAVED`. The Crux comparison `RECORD`: 25 percent against 63 percent raw became about 27 against 25 once the same containers, sources, week, and horizon were used. That is the parity gate and the reconcile-before-compare rule; both are step one of any scorecard and both are in Part 6.

## 5.6 Turn models into plans people can carry out

**The deliverable.** A model produces a number. A plan is what changes on Monday at which building, who does it, how they know it worked, and how they undo it. The handoff artifact is a one-page per site, and the partnership work is making sure each partner's piece exists before the date: the data scientist's model is in production with a

data contract, the engineer's configuration change is scheduled, finance's plan of record is updated, and the field has the page.

| PARTNER | WHAT THEY NEED FROM YOU | WHAT YOU NEED FROM THEM | THE ARTIFACT |
|---|---|---|---|
| Data scientists | The decision the model serves, the unit of work, the metric it must move, the baseline it is scored against, the horizon it is scored at (T-13 for a plan, T-1 for a flash; score each at its own horizon) | A model with a data contract: inputs named, refresh cadence, parity test against the ledger, a change log; a measured error, not a claimed one | Model card: purpose, inputs, error at horizon, owner, last parity date |
| Engineers | The configuration change in their terms: which lane, which pull time, which hold rule, which threshold; the date; the rollback | The change path: who can set a pull time, how long it takes, what tests it, and a change calendar so two changes do not land on one sort center in one week | Change ticket with the one-page attached and a rollback that is a configuration, not a hope |
| Finance | The saving by line with its rates and counts; the accrual treatment; the plan-of-record update so next week's variance does not show the change as a miss | The discount rate, the depreciation treatment, the plan of record, the invoice true-ups | Plan-of-record change memo, one page, both signatures |
| Field operations | What changes on Monday, in their words; what to watch on the flash; who to call; what "undo" looks like | The things the model does not know: the door that floods, the DSP that is short on Mondays, the yard that cannot take two more trailers at 21:00 | The one-page per site, walked in person before the date |

**The one-page for DS-3, ring 1 of the direct-inject rollout**

| LINE | CONTENT |
|---|---|
| What changes | From Monday November 17, FC-A loads a direct trailer for DS-3 at the 03:00 and 15:00 CPTs on door 22. DS-3's packages no longer ride through SC-1. |
| Why | DS-3 receives about 1,500 packages a day from FC-A; direct is $0.75 a package cheaper and 7.5 hours shorter. Expected: $1,125 a day; the DS-3 inbound cutoff moves from 06:30 to 05:00 with arrivals at 05:00 and 17:00. |
| Who does what | FC-A outbound: door 22 chute mapping and trailer schedule (ship lead). SC-1: DS-3 chute retired at the 01:00 window (sort lead). DS-3: inbound cutoff 05:00; 03:00 wave unchanged; afternoon wave takes the 17:00 arrival (station manager). Transportation: 2 tractors, 4 trailers on the lane (dispatch). |
| What to watch | Daily flash: direct trailer fill (plan 75%), on-time arrival at DS-3 (plan 97%), DS-3 rolled volume (plan ≤ 0.5%), SC-1 window completion. Weekly: pair TCPP against $1.81. |
| Rollback | Re-enable the DS-3 chute at SC-1 and remove door 22 from the schedule; one change ticket, pre-approved, 24 hours to take effect. Trigger: fill below 50% for five days or on-time below 95% for three. |
| Who to call | Named people with phone numbers, for each building, for the first two weeks. |

**Where it gets decided.** The change calendar is owned by the network design team and reviewed weekly with engineering; the one-page is signed by the three building leads. **The trap.** A model handed over as a spreadsheet, with the assumption that the field will translate it. They will not, and the measured result will be blamed on the model.

**The small version I have run.** The Freight Forwarding App rollout `RECORD-FROM-DOCS`: 905.9 engineer-months scoped, sequenced by partner priority, with 15 percent pre-launch testing held in the plan; import launch April 2022 with 11 partners in 10 countries. The Sephora architecture council `RECORD`: six dated sessions, two go-lives, escalations to zero; and the one-signed-available-for-sale rule per node per channel, with rollback triggers in the configuration. The Elementum ASN gate `RECORD`: a lane went live only at or above 95 percent valid data, which is the "what to watch" line with a number on it.

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## 5.7 Write recommendations senior leaders can decide on

**The deliverable.** A document that a senior leader can read in the time they have and make a decision from, where every number has a denominator and a source, the alternatives are present with the reason they lose, the risks come with triggers, and the ask is one sentence. The long form is the six-page narrative read in silence at the start of the meeting; the short form is the one-page that travels without you.

| SECTION | WHAT GOES IN IT | FROM THE SC-3 CASE |
|---|---|---|
| Recommendation | One sentence: do this, by when, for this much, to get this. | Approve $45 million for a 15,000-per-hour last-mile sort center at the north metro site, phase 1 live in 14 months, for $17.7 million a year at run rate and a day faster for 36,000 packages a day. |
| The decision asked | Exactly what the reader is approving today, and what comes back later. | Phase 1 capital and the site lease today; phase 2 returns at month six against the kill criteria. |
| The number and its denominator | The saving per package by path, the count, the net after second-order effects. | $0.55 on 40,000, $2.27 on 20,000, less $13,000 opex and $5,200 cannibalization a day. |
| What has to be true | The three assumptions the number depends on, with how they were tested. | Carrier share converts at 80% or better; direct-lane fill at 75%; labor at $22 and 1.15 paid-to-direct. |
| Alternatives considered | Each on the same basis, with the number that makes it lose. | Expand SC-1: saves $0.14 not $0.55, no speed. Renegotiate the carrier: best offer $4.10, saves $8,000 a day, no speed. Do nothing: $17.7 million a year. |
| Risks and triggers | What could go wrong, the number that would show it, the action. | Ramp below 80% of plan at month six: hold phase 2. Contribution below $0.40 measured: re-purpose as a cross-dock. |
| Appendix | The cash flows, the sizing, the inputs card, the model's parity record. | Three pages; nobody reads them in the room, and everyone checks them afterwards. |

**Rules for the page.** Lead with the recommendation, not the history. Every number is accompanied by its denominator and its status: measured, modeled, or assumed. Write the counter-argument yourself, in its strongest form, before the reader does. Numbers that could be added wrongly are kept in separate sentences. No internal system names in a page that will travel. **The trap.** A page that reads as advocacy: a leader who finds one unlabeled assumption stops trusting the rest, and the recommendation dies on a detail.

**The small version I have run.** The Flexport E-Team frame RECORD-FROM-DOCS: February 1, 2022, the visibility team was asked to publish its own review with arrival accuracy and interaction elimination first; the Crux memo that followed reported raw and normalized numbers side by side, 25 against 63 and then about 27 against 25, and recommended the decommission that ran August to October 2022 on 79 customers. The Sephora cadence change RECORD, six-week to four-week releases with defects down 50 percent, was decided from a one-page with a before and after and a trigger. The discipline this packet follows, labels on every number and nouns kept apart, is the same discipline.

## 5.8 Mentor and raise the bar: the modeling standards

**The deliverable.** A written standard that every model on the team meets before its number leaves the team, and a routine that teaches it. The standard is short enough to check in a review and specific enough to fail.

| # | STANDARD | THE CHECK | WHY IT EXISTS |
|---|---|---|---|
| 1 | Every number has a denominator, a unit, and a time grain | "$2.73 per package attempted, week 45, region" passes; "$2.73" fails | Most scorecard arguments are two people using different denominators |
| 2 | Anything with a cutoff is modeled as a distribution | Transit, pack cycle, unload time: a mean and a p90 at least; a plan built on the mean fails review | Cutoffs are met by the tail, not the average (3.1) |
| 3 | Reconcile before you compare | Same packages, same week, same sources, same horizon, or the comparison is not made; parity to the ledger within 1% before publication | The 25-against-63 that was really 27-against-25 |
| 4 | Baseline and counterfactual stated | What the packages would have cost on the alternative path, not "nothing" | Gross savings that are not net (5.1, 5.2) |
| 5 | Inputs card separate from outputs | One table of every assumed input with its source and status; the model reads from it; no hard-coded numbers | So a reviewer can change one input and see the output move (8.2 is this packet's) |
| 6 | Sensitivity on the top three inputs | Shown on the page, with the combined case | Finance asks; better to have answered |
| 7 | Second-order effects have their own lines | Cannibalization, fixed-cost dilution, door and yard capacity, fill on the new lane | Nets that survive the second meeting |
| 8 | Versioned, with a change log and a parity date | A model with no last-parity date is not in production | Drift is silent |
| 9 | Status labels: measured, modeled, assumed, public | On every number in every page that leaves the team | Credibility is lost once, on one number |
| 10 | The one-sentence test | The author can say the result in one sentence, with the denominator, to someone outside the team, and that person can repeat it | If it cannot be said, it is not understood |

**The routine.** A weekly model review, forty-five minutes, one model, the author presents the inputs card first and the result last; the reviewers' job is to attack the denominators and the counterfactual. A pairing rule: no number goes to a senior leader without a second person having recomputed it from the inputs card. A monthly read of one model that went wrong, with the standard it would have failed. For a new analyst: the first assignment is to reproduce an existing model's number from its inputs card without the author's help; what they cannot reproduce is what the standard is for. **The trap.** A standard that lives in a document nobody checks against. The check has to be in the review, every week, on a real model.

**The small version I have run.** At Flexport the seven frozen definitions and the parity gate became the team's standard for any number on the review, and the review time falling from 30 to 6 hours a week `RECORD-FROM-DOCS` is what a standard buys. At Sephora the 836-row log with 714 clean entries, 85.4 percent `RECORD`, was the artifact a new engineer learned the program from. I have not managed analysts as direct reports since MetricStream; the mentoring on this record is the matrix kind, running the review, pairing on the numbers, and making the standard the thing people are judged against.

# Part 6 — The supply chain scorecard, week over week

The scorecard is one page, published Monday morning for the week that ended Saturday, read in a forty-five-minute review, and built so that every movement on it can be traced to an input a building can move the following week. This part gives the metric tree, the ranking with the arithmetic behind it, the build rules, the week-over-week bridge worked on the reference network's week 44 to week 45, the page itself, and the way the review runs. The numbers are the reference network ILLUSTRATIVE; the build rules are the ones used on the record RECORD, cited in 6.3.

## 6.1 The metric tree: outputs, drivers, inputs, with definitions and owners

Three tiers. Outputs are what the customer and the P&L feel; they are reported but cannot be managed directly. Drivers are the denominators and shares that the outputs decompose into arithmetically. Inputs are the daily operating measures a single building owns, and each is tied to the driver it moves. Every line has one owner, one source, one grain, and a plan.

```
THE TREE                                                                    [illustrative v

 OUTPUTS   Transportation cost per package $2.834 (plan 2.710)     Delivery estimate accuracy 97.7% (plan
           Dock-to-door dollars vs plan +$93,677 / week              Volume vs plan +3.9%
                 |                                                         |
 DRIVERS   path mix (carrier share 17.0%, plan 14.6%)              exception shares: rolled 0.6, late lin
           trailer fill (SC-1 lanes 76%, plan 80)                    missed CPT 0.8, missort 0.45, carrie
           packages per route (190, plan 200)                      share of each losing the day (3.7)
           sort-center volume vs fixed cost (108k, plan 104k)
           missort rate (0.45%, plan 0.30)
                 |                                                         |
 INPUTS    pack-complete slack to CPT (h)      door utilization in the window     stops per hour on the r
           CPT compliance (%)                  window completion (%)              routes dispatched vs pl
           linehaul on-time arrival (%)        induct rate vs window              DSP and Flex capacity o
           yard dwell (min)                    sort-code mapping changes          failed first attempt (%
```

| TIER | METRIC | DEFINITION (NUMERATOR ÷ DENOMINATOR) | GRAIN | OWNER | SOURCE | PLAN |
|---|---|---|---|---|---|---|
| O | Transportation cost per package | Dock-to-door cost (5.5) ÷ packages attempted | Week; region; path | Network design (this role) | Ledger + delivery system | $2.710 W45 |
| O | Delivery estimate accuracy | Packages attempted on or before the promised date ÷ packages promised | Day; week; region | Regional operations | Delivery system | ≥ 97.5% |
| O | Dollars vs plan | (Actual rate − plan rate) × actual packages, by path, summed | Week; region | Transportation finance | Ledger | $0 |
| O | Volume vs plan | Packages attempted ÷ planned packages − 1 | Week; region; path | Planning | Delivery system + plan | ±3% noise band |
| D | Path mix | Packages on each path ÷ total packages | Week; region | Network design | Manifest | SC-1 44.8 / SC-2 25.6 / direct 15.0 / carrier 14.6 |
| D | Trailer fill at close | Positions used ÷ positions available (cube and weight beside it) | Day; lane | Transportation | Trailer close scans | 80% sort-center lanes; 75% direct |

| | | | | | | |
|---|---|---|---|---|---|---|
| D | Packages per route | Packages dispatched ÷ routes dispatched | Day; station | Station + DSP | Route system | 200 |
| D | Sort-center volume vs fixed | Packages sorted; fixed cost ÷ packages sorted | Week; site | Sort-center GM | Induct scans + ledger | 104,000 a day; $0.125 |
| D | Missort rate | Packages at the wrong station or route ÷ packages sorted | Day; site | Sort-center GM | Station induct scans | ≤ 0.30% |
| D | Exception shares | Packages in each exception stock ÷ packages, with the share of each that lost the day | Day; region | Network design | Scan events | Per 3.7 |
| I | Pack-complete slack | CPT − time the last package for the CPT left pack, in hours | Per CPT; FC | FC outbound | Pack scans | ≥ 1.5 h |
| I | CPT compliance | Trailers departed within 15 min of pull time ÷ scheduled | Day; FC | FC outbound | Gate scans | ≥ 98% |
| I | Linehaul on-time arrival | Arrivals before (window close − unload − sort) ÷ due | Day; lane | Transportation | Gate scans | ≥ 97% |
| I | Yard dwell | Minutes from gate-in to door, median and p90 | Day; site | Site GM | Yard system | p90 ≤ 30 min |
| I | Door utilization in the window | Trailer-hours at doors ÷ (doors × window hours) | Window; site | Sort-center GM | Door events | ≤ 85% |
| I | Window completion | Packages containerized before window close ÷ received for the window | Window; site | Sort-center GM | Container scans | ≥ 99% |
| I | Routes dispatched vs planned | Routes that left ÷ routes the plan called for | Day; station | Station manager | Route system | 100% |
| I | Rolled volume | Packages received for a wave not dispatched in it ÷ received | Day; station | Station manager | Station scans | ≤ 0.5% |
| I | Stops per hour | Stops completed ÷ on-road hours | Day; station | DSP | Route system | 20 urban; 10 rural |
| I | Failed first attempt | Stops not delivered on the first attempt ÷ stops | Day; station | Station manager | Delivery scans | ≤ 1.5% |

## 6.2 Which metrics matter most, ranked, with the elasticity behind the ranking

"Matters most" has two meanings and the scorecard needs both: dollars per week for a move of the size a normal week produces, and points of delivery estimate accuracy for the same. The elasticities below are computed on the reference network's week 45 (756,000 packages; own-network 627,500; sort-center-lane packages 517,300).

**Cost elasticities, dollars per week on the region**

| RANK | DRIVER | A NORMAL WEEK'S MOVE | ARITHMETIC | DOLLARS PER WEEK |
|---|---|---|---|---|
| 1 | Packages per route | ±10 (200 → 190) | ($264 ÷ 190 − $264 ÷ 200) × 627,500 own-network packages = $0.0695 × 627,500 | $43,600 |
| 2 | Path mix: carrier share | ±1 point | ($4.50 − $2.41 blended own-network rate) × 1% × 756,000 | $15,800 per point; the W45 move was 2.4 points, $38,000 (the mix term in 6.4) |

| | | | | |
|---|---|---|---|---|
| 3 | Trailer fill, sort-center lanes | ±3 points (80 → 77) | Per point: $1,024 ÷ (2,000 × 0.79) − $1,024 ÷ 1,600 = $0.0081 on 329,000 SC-1-path packages = $2,700; $0.0068 on 188,000 SC-2-path packages = $1,300; about $4,000 per point | $12,000 for 3 points; the W45 move was 4 points on SC-1 lanes only, $11,100 |
| 4 | Sort-center volume vs fixed cost | ±8% (a plan miss, not a weekly swing) | $13,000 ÷ 100,000 − $13,000 ÷ 108,000 = $0.0096 × 329,000 SC-1-path packages | $3,200, and it is a planning number, not an operating one; it tells you whether the plan was right |
| 5 | Missort | ±0.15 point (0.30 → 0.45) | 0.0015 × $2.00 × 517,300 sort-center-lane packages | $1,600. Small in dollars; large in speed (below) |
| 6 | Rolled at stations | ±0.2 point | 0.002 × 627,500 × $0.58 Flex premium | $700. Nearly free in dollars; the most expensive line in speed |

**Speed elasticities, points of delivery estimate accuracy per week**

| RANK | INPUT | A NORMAL WEEK'S MOVE | ARITHMETIC (SHARE × PROBABILITY OF LOSING THE DAY) | POINTS OF DEA |
|---|---|---|---|---|
| 1 | Rolled volume | +0.2 point | 0.2 × 100% lose the day | −0.20 |
| 2 | Linehaul on-time arrival | −1 point (one more trailer in 100 late) | 1 point of trailers ≈ 1 point of packages × 50% lose the day | −0.50 per point of arrival; a normal week moves it 0.3 points: −0.15 |
| 3 | CPT compliance | −0.5 point | 0.5 × 40% lose the day | −0.20 |
| 4 | Missort | +0.15 point | 0.15 × 70% | −0.11 |
| 5 | Carrier share | +1 point | 1 × 3% carrier late rate | −0.03 (but it is the cost driver ranked 2nd) |

**So the ranking.** On cost: packages per route first, path mix second, fill third; everything else is a plan check or a quality line. On speed: rolled volume first, on-time arrival second, CPT compliance third. Two inputs are leading indicators and get watched daily even though they carry no dollars: pack-complete slack (it predicts tomorrow's CPT misses) and door utilization in the window (it predicts the roll before the window closes). The two lines that look big and are not: sort-center volume against fixed cost, which is a planning variance, and missort dollars, which are small; missort matters for the speed column, not the cost one.

**SAY THIS — 120 WORDS**

The metrics that matter most are the ones with the biggest denominator effect for a move the network actually makes in a week. On cost that is packages per route, then how much volume we hand to carriers, then trailer fill; ten packages a route is worth forty thousand dollars a week on one region, a point of carrier share fifteen thousand, a point of fill about four thousand. On speed it is rolled volume, then linehaul on-time arrival, then pull-time compliance, because those are the exceptions most likely to lose a day. And I watch two leading inputs daily that carry no dollars: pack-complete slack and door utilization in the window, because they tell me about tomorrow's misses today.

## 6.3 Building the scorecard: frozen definitions, denominators, parity, review rules

| # | RULE | HOW IT IS DONE | WHERE IT CAME FROM |
|---|---|---|---|
| 1 | Freeze the definitions before the first page | One session with finance, operations, and the data owner. Each metric gets numerator, denominator, grain, source, owner, plan. Signed. A change is a dated version with the old and new values shown side by side for four weeks. | Flexport review: seven definitions frozen first RECORD |
| 2 | One denominator per metric, written on the page | Packages attempted for cost; packages promised for accuracy; trailers scheduled for compliance. The denominator is printed in the column header, not in a footnote. | The 5.5 definition table |
| 3 | Parity gate before publication | Weekly dollars tie to the ledger within 1%; package counts tie to the delivery system within 0.5%; trailer counts tie to gate scans exactly. The gate is run every week, not once; a failed gate publishes the page with the line marked "unreconciled" and no cause attached. | Flexport: parity within 1% against the source RECORD |
| 4 | Reconcile before you compare | Week over week and against plan only on the same package population, the same sources, and the same horizon. A plan set at T-13 is compared at T-13 accuracy; a flash at T-1 is a different line. | Crux: 25 against 63 became 27 against 25 RECORD |
| 5 | Plan by path, not blended | The plan is loaded as volume and rate for each path. The blended plan is computed, never typed. This is what makes the mix variance visible. | 6.4 |
| 6 | The five-percent rule | Any line that moves more than 5% week over week, or sits more than 5% from plan, carries a one-line cause and an owner on the page. "Unexplained" is an allowed cause for one week; in the second week it must have a name. | Review practice |
| 7 | Six weeks of trend, three weeks of sign | Six weekly columns on every line. Three consecutive moves in the same direction trigger a deep dive even if each is under 5%; that is a trend, and trends are cheaper to fix early. | Review practice |
| 8 | Volume, rate, mix on every dollar line | Every cost variance is split into the three terms before the review. A rate variance gets a building's name; a mix variance gets a network-design name; a volume variance gets planning's name. | 6.4 |
| 9 | Two columns for dollars: accrued and trued-up | Carrier and DSP invoices lag. The page shows the accrued rate and, four weeks later, the trued-up rate in a second column so a late invoice is never read as an operating change. | 5.5 accrual rule |
| 10 | Actions with dates, carried forward | The bottom of the page lists open actions from prior weeks with owner and due date. An action closes only when its metric moved. | Review practice |
| 11 | Published at a fixed time, in a fixed shape | Monday 10:00 for Sunday to Saturday. Same lines, same order, every week. Readers learn where to look; movement is visible because the shape does not move. | Review practice |
| 12 | Hours to produce are a metric too | The scorecard's own build time is tracked. If it takes more than four hours a week, the sources are wrong, not the people. | Flexport: 30 to 6 hours a week RECORD |

## 6.4 Week-over-week math: the variance and the volume-rate-mix bridge, worked

The region's blended transportation cost per package rose from $2.7135 in week 44 to $2.8339 in week 45, up 12.0 cents, 4.4%. The bridge says where. Shares are path mix; rates are cost per package on each path.

**Inputs: week 44 and week 45 by path**

| PATH | W44 SHARE | W44 RATE | W45 SHARE | W45 RATE | WHAT CHANGED ON THE PATH |
|---|---|---|---|---|---|
| Via SC-1 | 44.80% | $2.5689 | 43.54% | $2.6654 | Fill on SC-1 lanes 80% → 76%; packages per route 200 → 190; missort 0.30% → 0.45%; SC-1 volume 100,000 → 108,000 a day |
| Via SC-2 | 25.60% | $2.4760 | 24.88% | $2.5485 | Packages per route 200 → 190; missort 0.30% → 0.45% |
| Direct | 15.00% | $1.8120 | 14.58% | $1.8815 | Packages per route 200 → 190 |
| Carrier | 14.60% | $4.5000 | 17.00% | $4.5000 | Share up 2.4 points: a same-day site's overflow was handed to carriers in W45 |
| Blended | 100% | $2.7135 | 100% | $2.8339 | Volume 100,000 → 108,000 a day; 700,000 → 756,000 a week |

**The bridge**

| STEP | TERM | FORMULA → SUBSTITUTION | RESULT |
|---|---|---|---|
| 1 | Total change | $2.8339 − $2.7135 | +$0.1204 per package; × 756,000 = $91,000 for the week |
| 2 | Mix effect | Σ (W45 share − W44 share) × W44 rate = (−0.0126 × 2.5689) + (−0.0072 × 2.4760) + (−0.0042 × 1.8120) + (+0.0240 × 4.5000) | +$0.0502. Carrier share alone is +$0.108; the other three paths shrinking offsets −$0.058 |
| 3 | Rate effect | Σ W45 share × (W45 rate − W44 rate) = 0.4354 × 0.0965 + 0.2488 × 0.0725 + 0.1458 × 0.0695 + 0.17 × 0 | +$0.0702 |
| 4 | Check | 0.0502 + 0.0702 | $0.1204. Ties. |
| 5 | Rate effect, decomposed: fill | 0.4354 × [($640 + $384) ÷ (2,000 × 0.76) − ($640 + $384) ÷ (2,000 × 0.80)] = 0.4354 × $0.0337 | +$0.0147 |
| 6 | Rate effect: route density | (0.4354 + 0.2488 + 0.1458) × ($264 ÷ 190 − $264 ÷ 200) = 0.83 × $0.0695 | +$0.0577 |
| 7 | Rate effect: missort | (0.4354 + 0.2488) × (0.0045 − 0.0030) × $2.00 | +$0.0021 |
| 8 | Rate effect: fixed-cost dilution at SC-1 | 0.4354 × ($13,000 ÷ 108,000 − $13,000 ÷ 100,000) | −$0.0042 |
| 9 | Check | 0.0147 + 0.0577 + 0.0021 − 0.0042 | $0.0703. Ties to step 3 within rounding. |
| 10 | Against plan | Plan W45 $2.710 (volume 104,000 a day); actual $2.8339; rate variance $0.1239 × 756,000; volume variance +4,000 a day, +3.85% | +$93,677 unfavorable on rate; volume inside the noise band |

**What the bridge says to do.** Route density is 48% of the move and belongs to the stations: find out whether it is the DSP capacity (routes dispatched vs planned), the catchment (stops per hour), or the wave timing. Carrier share is 42% and belongs to network design: the same-day overflow rule that sent 2.4 points to carriers needs a cost check against an afternoon own-network wave. Fill is 12% and belongs to transportation on the SC-1 lanes:

four points in one week on one lane group is a schedule change or a volume shift, and the daily flash will say which day it started. Missort is small in dollars and goes to the speed column. The fixed-cost dilution is good news and is said in one clause.

---

## 6.5 The one-page scorecard, week 45 against week 44 and against plan

This is what the page looks like. Six weeks of trend, plan, week-over-week, and a status that follows the five-percent rule: green inside 2% of plan, amber within 5%, red beyond. Causes are one line and name an owner.

| LINE | PLAN | W40 | W41 | W42 | W43 | W44 | W45 | WOW | VS PLAN | CAUSE, OWNER |
|---|---|---|---|---|---|---|---|---|---|---|
| Packages attempted (000s/wk) | 728 | 686 | 693 | 697 | 704 | 700 | 756 | +8.0% | +3.9% | Promotion week; inside the band on the re-forecast. Planning. |
| **Transportation cost per package** | 2.710 | 2.702 | 2.698 | 2.711 | 2.719 | 2.714 | 2.834 | +4.4% | +4.6% | Mix +5.0¢ (carrier share), route density +5.8¢, fill +1.5¢. Three owners below. |
| Dollars vs plan ($000s/wk) | 0 | −5 | −8 | +1 | +6 | +3 | +94 | | | Rate variance; volume inside the band. |
| Rate: via SC-1 | 2.560 | 2.561 | 2.558 | 2.566 | 2.572 | 2.569 | 2.665 | +3.8% | +4.1% | Fill 76% on SC-1 lanes from Tuesday; route density. Transportation; stations. |
| Rate: via SC-2 | 2.470 | 2.474 | 2.471 | 2.477 | 2.480 | 2.476 | 2.549 | +2.9% | +3.2% | Route density only. |
| Rate: direct | 1.812 | 1.815 | 1.810 | 1.818 | 1.814 | 1.812 | 1.882 | +3.8% | +3.8% | Route density only. |
| Rate: carrier | 4.500 | 4.500 | 4.500 | 4.500 | 4.500 | 4.500 | 4.500 | 0 | 0 | Rate card; accrued. W42 true-up +$14k posted this week. |
| Carrier share | 14.6% | 14.4 | 14.7 | 14.5 | 14.8 | 14.6 | 17.0 | +2.4 pt | +2.4 pt | Same-day site overflow handed to carriers Wed–Sat. Network design: afternoon own-network wave costed by Friday. |
| Fill, sort-center lanes | 80% | 80.4 | 79.8 | 80.1 | 79.6 | 80.0 | 76.9 | −3.1 pt | −3.1 pt | SC-1 lanes 76%; a 7th daily departure added Tuesday for the promotion and not removed. Transportation. |
| Packages per route | 200 | 201 | 199 | 200 | 198 | 200 | 190 | −5.0% | −5.0% | Routes dispatched 104% of plan on 8% more volume: stations added routes faster than packages. Stations; DSP route plan. |
| Sort-center volume, SC-1 (000s/day) | 104 | 98 | 99 | 100 | 101 | 100 | 108 | +8.0% | +3.8% | Fixed-cost dilution −0.4¢. Good. No action. |
| Missort rate | 0.30% | 0.29 | 0.31 | 0.30 | 0.32 | 0.30 | 0.45 | +0.15 pt | +0.15 pt | Sort-code remap for the DS-3 direct lane left two ZIPs on the old chute. SC-1; fixed Thursday, confirm this week. |
| **Delivery estimate accuracy** | 97.5% | 97.9 | 97.8 | 98.0 | 97.9 | 97.9 | 97.7 | −0.2 pt | +0.2 pt | Above plan, down 0.2: missort −0.1, carrier mix −0.07. Rolled flat but rising three weeks. |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Rolled volume | 0.5% | 0.4 | 0.4 | 0.4 | 0.5 | 0.55 | 0.6 | +0.05 pt | +0.1 pt | Three weeks same sign: deep dive opened. Two stations short of vans on Mondays. Stations. |
| Linehaul on-time arrival | 97.0% | 97.2 | 96.9 | 97.1 | 96.8 | 96.3 | 96.1 | −0.2 pt | −0.9 pt | SC-1 01:00 window door queue, 4 trailers Tue and Thu. Crew-of-3 trial from Monday. SC-1. |
| CPT compliance | 98.0% | 98.3 | 98.1 | 98.4 | 98.0 | 98.1 | 98.0 | −0.1 pt | 0 | — |
| Window completion, SC-1 | 99.0% | 99.2 | 99.1 | 99.0 | 98.9 | 98.7 | 98.5 | −0.2 pt | −0.5 pt | Same cause as on-time arrival; same action. |
| Pack-complete slack, FC-A (h) | ≥1.5 | 1.9 | 1.8 | 1.9 | 1.6 | 1.7 | 1.3 | −0.4 | −0.2 | Leading indicator. Promotion volume at the 21:00 CPT. Watch daily; flex labor from the 03:00 CPT. FC-A. |
| **Open actions.** (1) Cost an afternoon own-network wave for same-day overflow vs carrier handoff; network design; Friday. (2) Remove or fill the seventh SC-1 departure; transportation; Wednesday. (3) Route plan vs packages at the two stations adding routes ahead of volume; stations; Thursday. (4) Confirm the DS-3 sort-code fix held: missort back under 0.30% by W46. (5) Rolled-volume deep dive, three weeks same sign; stations; read-out next review. (6) Crew-of-3 trial at SC-1's 01:00 window; SC-1; read-out next review. Carried from W44: DSP Monday van capacity at DS-7 and DS-12, open. | | | | | | | | | | |

## 6.6 Running the review: agenda, escalation grammar, the actions that come out

| MINUTES | SEGMENT | WHAT HAPPENS |
|---|---|---|
| 0–5 | Read in silence | The page and the bridge. Nobody presents the numbers; everyone has read them. |
| 5–10 | Parity and definitions | Did the gate pass; any definition change this week, shown side by side. If the gate failed, which lines are unreconciled. No causes are discussed on an unreconciled line. |
| 10–30 | Red lines, by dollars then by points | Each red line in order of size, in the escalation grammar below. The owner speaks; the room asks only about the cause and the action. Three minutes a line. |
| 30–38 | Amber and three-week trends | Lines under 5% that have moved the same way three weeks; deep dives opened or read out. |
| 38–43 | Open actions | Each carried action: closed because the metric moved, or still open with a new date and the reason. |
| 43–45 | The one sentence | The chair states the week in one sentence with its number and denominator, and it goes at the top of next week's page. |

**The escalation grammar: one line, seven slots**

| METRIC | PLAN | ACTUAL | CAUSE (ONE CLAUSE, A THING THAT HAPPENED) | ACTION (A VERB AND AN OBJECT) | OWNER (A NAME) | DATE |
|---|---|---|---|---|---|---|
| Fill, SC-1 lanes | 80% | 76% | A seventh daily departure added Tuesday for promotion volume and not removed when volume normalized | Remove the seventh departure or fill it from the 03:00 CPT's early work | Lane manager, SC-1 lanes | Wednesday |
| Packages per route | 200 | 190 | Two stations dispatched routes at 104% of plan on 108% volume | Re-cut the route plan to the package forecast, not the route count | Station managers, DS-7 and DS-12 | Thursday |

> Things that are not causes: "volume," "peak," "the system," "the DSP." A cause is an event with a day attached. Things that are not actions: "monitor," "investigate," "align." An action changes a configuration, a schedule, a staffing plan, or a rule, and the metric that proves it is named.

**What comes out.** An action list with names and dates; at most two deep dives open at once; a one-sentence summary that travels to the senior leadership review; and, once a quarter, a definitions review where the only question is whether any line should be retired or added. The review is not where the model is argued. The model is argued in the modeling review (5.8); the scorecard review takes the numbers as given and asks what to do about them.

**SAY THIS — 135 WORDS**

I build the scorecard before I run it: definitions frozen and signed, one denominator per line printed in the header, a parity gate to the ledger every week, and the plan loaded by path so mix cannot hide in rate. Then the page is six weeks of trend, plan, week over week, and a status, with the volume-rate-mix bridge underneath every dollar line. The review reads in silence, takes red lines by size in a fixed grammar, metric, plan, actual, cause, action, owner, date, and closes an action only when the metric moves. Three weeks of the same sign opens a deep dive even if no week was red. The build is the discipline I used at Flexport: seven definitions frozen, parity within one percent, and the review time fell from thirty hours to six.

# Part 7 — Your record, mapped to this job

The job is network design at a scale you have not run. The record is the same methods at program scale: measure at the unit of work, reconcile before comparing, change one rule with a gate, and build the review that proves it. This part puts each responsibility next to the story that shows the method, with the number you may say and the status it carries from the latest Amazon and Apple packets. Statuses: RECORD the reconciled career fact; USER ANCHOR carried by the packets and flagged to confirm; ILLUSTRATIVE a worksheet input or design case. Career numbers keep their nouns and are never added to each other.

## 7.1 Story to responsibility map, with the numbers you may say and their status

| RESPONSIBILITY | STORY | WHAT YOU DID, IN THE JOB'S TERMS | NUMBERS YOU MAY SAY | STATUS |
|---|---|---|---|---|
| Design and roll out a new network model; prove cost and speed | Elementum lane program, 2017–2019 | Baselined 200 lanes at the shipment level; found lead-time variance was most of the buffer on the worked lane; re-set the rule from the measured distribution; released inventory at the same service level. Same packages, one rule changed, measured on the same lanes. | 200 lanes; about 580,000 units, about $1.5 million one time, at 99 percent service. Worked lane: brochure 18 days, mean 21, p99 34, about 400 shipments. | record. Lane A's 12,937 → 8,102 and the 97.5 percent variance share are illustrative lanes that tie to the record net. |
| Profitability of sortation capacity; design standards | Flexport cost per container; Elementum spend cut | Built the variable cost from the ledger's own rates, labeled the assumed rates, reconciled to the endpoint finance already held; ranked the work by spend share and did it in cushion order. | $0.66 to $0.03 a container, about 95 percent. 597 lanes as 80 percent of spend. | record endpoints for the container cost; the $1.50/$120 and $0.80/$150 worksheet rates are illustrative. Spend cut is a user anchor: say "in my earlier record." |
| Capital plans and business cases, defended with finance | Flexport roadmap re-sequencing; Elementum Austria | Re-sequenced a roadmap so value landed in month six not month eighteen; priced the avoidance with a stated model; executed a per-site saving and multiplied by a confirmed site count; pre-wired the owners of each count. | Month 18 to month 6; 66 percent less effort; about $5 million avoided once. Austria about $500,000 a year; about $8 million across 25 sites. | record for the roadmap and the Austria execution. The $2.85 million + $1.52 million model under the $5 million is modeled. The $5 million Austria base is a user anchor. Avoidance, not annual. |

| | | | | |
|---|---|---|---|---|
| Design and run dispatch pilots; scale what works | Sephora Utah robotics gate; Flexport wave hold; Sephora available-for-sale | Pre-registered a bar, throttled volume instead of switching, scaled on the number; held a release that missed its gate; put rollback triggers in the configuration. | 98 percent bar; week two 94 percent; about 30 percent held back; full in week three. Wave held at 76 against 80. Oversell recurring to zero after June 5, 2025. | record. No baseline incident count for oversell; in-transit sigma of 0.3 day is a user anchor. |
| Define, track, report cost per package and performance vs plan | Flexport weekly review; Crux reconciliation | Froze seven definitions before publishing; parity to the source within one percent; normalized a comparison to the same containers, sources, week, and horizon before reporting it; scored forecasts at their horizon. | Seven definitions; system accuracy 60 to 90 percent; parity within 1 percent; review 30 to 6 hours a week; $103,680 priced at an assumed $90. Crux 25 against 63 raw, about 27 against 25 normalized; 90 percent against an 80 north star after sources. | record; the $103,680 is record-from-docs arithmetic, priced not saved; the Crux populations (1,000 / 400 / 252 / 270 / 250) are illustrative. |
| Turn models into plans people can carry out | Flexport Freight Forwarding App; Sephora architecture council; Elementum ASN gate | Scoped and sequenced a multi-team build with testing held in the plan; ran a dated council that took escalations to zero; set a go-live gate on data quality that a lane had to pass. | 905.9 engineer-months, 75.5 engineer-years; P0 14 percent; +15 percent pre-launch testing; import launch April 2022, 11 partners, 10 countries. Council dates March 29 to August 14, two go-lives. ASN gate at or above 95 percent valid. | record-from-docs for the FFA scope; record for the council and the gate. The 480/800 and 760/800 ASN volumes are illustrative. |
| Write recommendations senior leaders can decide on | Flexport E-Team frame and Crux memo; Sephora cadence change | Published raw and normalized numbers side by side and recommended a decommission that ran; moved a release cadence on a one-page with a before, an after, and a trigger. | February 1, 2022 frame; cancellations August 1, September 1, October 24, 2022; 79 customers. Cadence six to four weeks; defects down 50 percent. | record-from-docs for the frame and dates; record for the cadence. Cost to run Crux not on record. |
| Mentor; raise the bar on modeling standards | Flexport review standard; Sephora 836-row log | Made frozen definitions and a parity gate the team's standard for any number on the review; left an artifact a new engineer could learn the program from. | 30 to 6 hours a week. 714 of 836 rows clean, 85.4 percent; efficiency plus 15 percent; zero customer incidents. | record. The 15 points have no before/after rates on record. Direct reports only at MetricStream; since then, matrix. |
| Peak and capacity under load (asked in every transportation loop) | Manhattan Associates | Allocation and order promising across three distribution centers and a hundred stores on the peak day; fill rate above goal. | 3 DCs, 100 stores; 99 percent fill against a 95 percent goal. | record. Introduction only; do not build a story on it. |

| | | | | |
|---|---|---|---|---|
| Network design cases | Walmart promise; Google Fiber spine | Design exercises in the master packet; not delivered work. | None. | design case, illustrative model only. Say so if raised. |

**What is not on the record, said plainly if asked.** You have not sized a sortation building, owned a linehaul rate card, or run a capital case through a capital committee. You have built per-unit cost models from ledgers, re-set rules from measured distributions, run gated pilots, and built the review that proved them, each at program scale. The honest sentence is: "The method is the same and I have run it end to end on lanes, containers, and releases; the objects here are trailers, windows, and routes, and the first month is learning their rates from the people who own them."

## 7.2 Sixty seconds on why you can do this job, and the thirty-second cut

**SAY THIS — 201 WORDS**

This job is three things I have done at program scale: measure a network at the unit of work, change one rule and prove it on the same units, and build the weekly page that keeps it honest. At Elementum I measured two hundred lanes at the shipment level, found that lead-time variance was most of the buffer, re-set the rule from the measured distribution, and released about five hundred eighty thousand units, about one and a half million dollars one time, at ninety-nine percent service. At Flexport I took cost per container from sixty-six cents to three, built the weekly review from seven frozen definitions with parity to the source within one percent, and when a comparison came in at twenty-five against sixty-three, normalized it to the same containers, sources, week, and horizon, where it was twenty-seven against twenty-five; that memo retired the product. At Sephora I gated a robotics rollout at ninety-eight percent, held thirty percent of volume back in week two, and went full in week three. Here the units are packages, trailers, windows, and routes. I would spend the first month learning their rates from the people who own them, and then run the same three moves.

**SAY THIS — 83 WORDS**

I measure at the unit of work, change one rule, and prove it on the same units with a page people trust. Elementum: two hundred lanes, buffers re-set from the measured distribution, about one and a half million released at ninety-nine percent service. Flexport: cost per container sixty-six cents to three, a weekly review built on seven frozen definitions and one-percent parity. Sephora: a rollout gated at ninety-eight percent, throttled, then scaled. Trailers, windows, and routes are new nouns for the same method.

## 7.3 Trade-offs you should expect to be asked

| THE QUESTION | THE ANSWER, WITH THE ARITHMETIC IT RESTS ON | WHERE IN THE PACKET |
|---|---|---|
| Fill versus speed: do you hold a trailer to fill it? | Only when the lane's measured slack covers the hold. On the reference lane the window carries 36 minutes; a 30-minute hold on a 70-percent trailer that saves a departure 60 percent of the time is worth about $106 a lane-day in expectation and is piloted with a kill switch on on-time arrival, not assumed. | 2.6, 3.1, 5.4 |

| | | |
|---|---|---|
| Hub versus direct: when does a direct lane win? | When the station takes more than about 300 packages per planned departure from the origin, because the middle mile through a sort center costs about $0.94 and a $288 trailer breaks even at 306. Above that, direct is cheaper and 7.5 hours shorter; below it the sort center's consolidation wins. | 5.1 |
| Build a sort center or hand the volume to a carrier? | Against a carrier every sort center wins; the real alternative is the neighboring sort center or a direct lane, and that cushion is usually under 2×. Decide on contribution against the real alternative plus the speed count, not on carrier savings. | 5.2 |
| Size for average or peak? | Peak window of the design year, at 85 percent: for SC-3, 141,000 on the peak day, 35,000 in the window, a 15,000-per-hour sorter and 18 doors, phased so year-one capital matches year-one volume. | 5.3 |
| More routes or denser routes? | Ten packages a route is about seven cents on every own-network package, $43,600 a week on one region; but a 260-package route misses the window. Density up to the window limit, then routes. | 3.5, 6.2 |
| Cut cost or protect the promise this week? | Read both columns. Rolled volume is nearly free in dollars and the most expensive line in speed; carrier share is the opposite. The scorecard shows them side by side so the trade is explicit, and the plan says which one the quarter is buying. | 6.2 |
| Model says yes, the field says no. | The field knows the yard that cannot take two more trailers at 21:00. Walk the one-page with them before the date; change the page where they are right; pilot where they are not sure. A model that cannot survive the dock lead's objection is not finished. | 5.6 |
| Why is your cost per package different from finance's? | Denominator or scope, every time. Attempted versus shipped; dock-to-door versus all-in; accrued versus invoiced. Reconcile before you compare: the 25-against-63 that was 27-against-25. | 5.5, 6.3 |
| What have you not done? | Sized a building, owned a rate card, or taken a case through a capital committee. Said plainly, followed by the method sentence in 7.1. | 7.1 |

# Part 8 — Formula sheet, inputs card, numbers card, public facts

## 8.1 Formula sheet: symbolic, substituted, computed

| QUANTITY | FORMULA | SUBSTITUTED ON THE REFERENCE NETWORK | RESULT |
|---|---|---|---|
| Stock balance | S(t+1) = S(t) + In(t) − Out(t) | Pack queue at 13:00 on the +60% day: 8,200 + 8,000 − 5,000 | 11,200 |
| Capacity | C = resources × rate × hours | Pack: 58 packers × 100 per hour × 10 h × 2 shifts ÷ 1.15 | About 100,000 a day |
| Utilization | ρ = demand ÷ capacity | SLAM at the peak hour: 7,000 ÷ 7,500 | 93% |
| Little's law | Time in queue = stock ÷ throughput | 2,500 in the pack queue ÷ 5,000 per hour | 30 min |
| Queue growth above the knee | Wait ∝ ρ ÷ (1 − ρ) × ($c^2_a$ + $c^2_s$) ÷ 2 × service time | From ρ = 0.85 to 0.93 the ρ ÷ (1 − ρ) term goes 5.7 → 13.3 | Waits more than double for 8 points of load |
| Drain time of a backlog | Backlog ÷ (capacity − arrivals) | 11,200 ÷ (5,000 − 0) after arrivals stop at 14:00 | 2.24 h; pack complete 16:14 |
| Labor hours | Units ÷ rate × (1 + indirect) | 160,000 ÷ 160 × 1.15 | 1,150 paid pick hours |
| Heads per shift | Paid hours ÷ shifts ÷ hours per shift | 1,150 ÷ 2 ÷ 10 | 58 pickers |
| Cost per package, one resource | Resource cost ÷ packages that rode it | $640 ÷ 1,600 | $0.40 |
| Cost per package, a path | Σ over legs of (resource cost ÷ packages) | 0.40 + 0.30 + 0.24 + 0.30 + 1.32 | $2.56 dock-to-door via SC-1 |
| Blended cost per package | Σ share × path cost | 0.448 × 2.56 + 0.256 × 2.47 + 0.15 × 1.81 + 0.146 × 4.50 | $2.708 |
| Trailer fill (positions) | Packages ÷ positions | 1,600 ÷ 2,000 | 80% |
| Roll risk at a plan fill | P(demand > positions) = 1 − Φ((positions − μ) ÷ σ) | μ = 1,800, σ = 180: z = 1.11 | 13.3% of trailers overflow at a 90% plan |
| Expected overflow | σ × [φ(z) − z × (1 − Φ(z))] | 180 × [0.215 − 1.11 × 0.133] | About 12 packages per trailer at a 90% plan |
| Transit quantile | $t_p$ = μ + $z_p$ × σ | 4.0 + 1.28 × 0.35 | p90 = 4.45 h |
| Window miss probability | P(transit > latest useful arrival − departure) | P(z > (4.6 − 4.0) ÷ 0.35 = 1.71) | 4.4% of trailers |
| Backward schedule | $Cutoff_{n-1}$ = $cutoff_n$ − $duration_n$ − $buffer_n$ | 09:00 − 2.5 → 06:30 − 2.5 → 04:00 − 3.0 → 01:00 − 4.0 → 21:00 − 2.0 → 19:00 − 2.0 | Customer cutoff 17:00 Monday for Tuesday delivery |
| Trailers in transit | Departure rate × transit | 28 ÷ 24 × 4.0 | 4.7 |
| Tractors for a lane | Departures ÷ (24 ÷ cycle) × (1 + spare) | 28 ÷ (24 ÷ 11) × 1.10 | About 15 |
| Door-slots in a window | Doors × window hours ÷ hours per trailer | 12 × 3 ÷ 2 | 18 slots for 16 trailers |
| Doors needed | Trailers × hours per trailer ÷ window hours ÷ 0.85 | 16 × 2 ÷ 3 ÷ 0.85 | 12.5, so 13 to 14 |

| | | | |
|---|---|---|---|
| Blended rate of serial touches | $1 \div \Sigma\ (1 \div \text{rate}_i)$ | 1 ÷ (1/400 + 1/1,000 + 1/600 + 1/600) | 146 packages per labor-hour |
| Fixed cost per package | Fixed cost ÷ packages | $13,000 ÷ 100,000; ÷ 60,000 | $0.13; $0.22 |
| Route cost per package | Route cost ÷ packages per route | $264 ÷ 200; ÷ 190 | $1.32; $1.39 |
| Route feasibility | Stops ÷ stops per hour ≤ window − breaks and return | 176 ÷ 20 = 8.8 h ≤ 11 − 2.2 | Feasible at 220 packages; not at 260 |
| Breakeven direct lane | q* = trailer cost ÷ middle-mile cost via the hub | $288 ÷ $0.94 | 306 packages per departure |
| Contribution of a sort center | (Alternative path cost − variable cost here) × packages − fixed | ($2.59 − $2.33) × 60,000 − $8,400 | $7,200 a day |
| Breakeven volume | Fixed ÷ contribution per package | $8,400 ÷ $0.26 | 32,308 a day |
| Net saving of a new site | Σ saving × packages − new fixed − cannibalization | $67,400 − $13,000 − $5,200 | $49,200 a day |
| NPV | $\Sigma\ CF_t \div (1 + r)^t$ − capex | $8.9 \div 1.1 + 17.7 \times \Sigma_{t=2..10}\ 1.1^{-t} - 45$ | About $56 million |
| Payback | Capex ÷ annual net saving | $45M ÷ $17.7M | 2.5 years |
| Pilot expected value | P(trigger) × [P(save) × gain − P(late) × loss] | 0.35 × [0.6 × 640 − 0.05 × 1,600] | $106 per lane-day |
| Sample size, two arms | $n = 2\ (z_{0.975} + z_{0.8})^2\ \sigma^2 \div \delta^2$ | $2 \times 2.8^2 \times 0.2^2 \div 0.05^2$ | 251 lane-days per arm |
| Difference in differences | $(T_a - T_b) - (C_a - C_b)$ | Treated fill +4.5; control +0.0 | +4.5 points attributable |
| Mix effect | $\Sigma\ (\text{share}_1 - \text{share}_0) \times \text{rate}_0$ | W44 → W45 | +$0.0502 |
| Rate effect | $\Sigma\ \text{share}_1 \times (\text{rate}_1 - \text{rate}_0)$ | W44 → W45 | +$0.0702 |
| Rate variance vs plan, dollars | (Actual rate − plan rate) × actual packages | $0.1239 × 756,000 | +$93,677 |
| Elasticity of a driver | Δ cost per package for a unit move × packages exposed | ($264 ÷ 190 − $264 ÷ 200) × 627,500 | $43,600 a week for 10 packages a route |
| Delivery estimate accuracy | 1 − Σ (exception share × P(loses the day)) | 1 − (0.32 + 0.60 + 0.32 + 0.60 + 0.51)% | 97.7% |

## 8.2 Reference-network inputs card: every illustrative input in one table

Every number in the worked examples derives from these inputs. Change one and the outputs move; none of them is an Amazon figure.

| INPUT | VALUE | INPUT | VALUE | INPUT | VALUE |
|---|---|---|---|---|---|
| FC-A packages a day | 100,000 | Units per package | 1.6 | Operating hours a day | 20 (2 × 10 h) |
| Pick rate | 160 units per hour | Pack rate | 100 packages per hour | Indirect-to-direct | 1.15 |

| | | | | | |
|---|---|---|---|---|---|
| Wage, all buildings | $22 per hour | Packing materials | $0.15 per package | SLAM | 3 lines × 2,500 per hour |
| Ship sorter | 10,000 per hour | Dock load rate | 500 per hour per loader | Peak-hour factor | 1.4 × average |
| Outbound doors, FC-A | 24 | CPTs a day | 4 (03:00, 09:00, 15:00, 21:00) | Dock buffer | 2.0 h before CPT |
| Order-to-pack cycle | mean 1.2 h; p90 2.0 h | Trailer positions | 2,000 | Usable cube | 3,000 ft³ |
| Package cube; weight | 0.6 ft³; 2.5 lb | Payload | 45,000 lb | Plan fill | 80% hub lanes; 75% direct |
| Fill standard deviation | 10% of mean | Linehaul rate | $3.20 per mile all-in | FC-A → SC-1 | 200 mi; 4.0 h; σ 0.35 h |
| FC-A → SC-2 | 150 mi; 3.0 h | FC-A → DS direct | 90 mi; 2.0 h; 5 stations | SC-1 → DS-7 | 120 mi; 2.5 h |
| Path shares from FC-A | 44.8 / 25.6 / 15.0 / 14.6 | Carrier rate, dock-to-door | $4.50 | Carrier pickup | 2,400 packages |
| SC-1 packages a day | 100,000 in 4 windows | Window | 3 h | SC-1 doors | 12 |
| Unload | 2 × 400 per hour per door | Induct | 1,000 per hour per station | Sorter, SC-1 | 20,000 per hour |
| Containerize; load | 600 per hour each | SC-1 fixed cost | $13,000 a day | SC-2 | 60,000 a day; $8,400 fixed |
| Missort rate; cost | 0.3%; $2.00 | DS-7 packages a day | 6,000 | DS-7 sort window | 04:00–09:00 |
| DS rates: unload, sort, stage | 600, 250, 400 per hour | DS-7 fixed cost | $560 a day | Packages per route | 200 |
| Packages per stop | 1.25 | Route length; window | 8 h; 09:00–20:00 | Route cost | $264 ($22 + $8 + $3 per hour) |
| Flex cost | $1.90 per package | Failed first attempt | 1.5% | Exception day-loss shares | CPT 40%, late 50%, missort 70%, rolled 100% |
| Carrier late rate | 3% | SC-3 capex; opex | $45M; $13,000 a day | Discount rate; horizon | 10%; 10 years; 50% year-one ramp |
| Growth; peak factor | 8% a year; 1.6 × average day | Pilot trigger / save / late | 35% / 60% / 5% | Fill σ across lane-days | 0.20 |
| W44 volume | 100,000 a day | W45 volume | 108,000 a day | W45 changes | fill 76% SC-1 lanes; 190 per route; missort 0.45%; carrier 17% |

## 8.3 Numbers card: career numbers used in this packet, with status

Carried from the Amazon team-match packet's numbers card and the Apple packet's chronology. Status is how you may say it. Nouns stay separate; nothing on this card is added to anything else on it.

| METRIC | VALUE | USED IN | STATUS |
|---|---|---|---|
| Tenure | Fifteen years | 7.2 | record |
| Chronology | Manhattan Associates (start); MetricStream; Elementum 2017–2019; Flexport 2021–2022 (visibility team from May 2020); Sephora April 2023–present | 7.1 | record, from the Apple packet's C2 |
| Manhattan peak | 3 DCs, 100 stores; 99 percent fill vs 95 percent goal | 7.1 | record; introduction only |
| Lanes measured | 200 lanes; worked lane brochure 18 days, mean 21, p99 34; about 400 shipments | 5.1, 7.1, 7.2 | record for the 200 and the 99 percent; the worked lane is the case |
| Program release | About 580,000 units, about $1.5 million, one time, 99 percent service | 5.1, 7.1, 7.2 | record. Not added to the $2 million forecast release or the $2 million freight saving. |
| Lane A buffer; variance share | 12,937 → 8,102; 97.5 percent of variance from lead time | 7.1 | illustrative lane tying to the record net |
| Spend cut | 597 lanes = 80 percent of spend | 5.2, 7.1 | user anchor. Say "in my earlier record." |
| Completeness gate | 60 to 95 percent; live at or above 95 percent valid ASN | 5.6, 7.1 | record; 480/800 and 760/800 volumes illustrative |
| Cost per container | $0.66 to $0.03; about 95 percent | 5.2, 7.1, 7.2 | record endpoints; worksheet rates illustrative; $0.18 per interaction is finance's rate, not the ticket price |
| Automation | About 40 to about 90 percent | not spoken here | record; the 60-to-97 worksheet is not blended with it |
| System accuracy and parity | 60 to 90 percent; parity within 1 percent | 5.5, 6.3, 7.1 | record. Not arrival accuracy. |
| Review hours | 30 to 6 hours a week; $103,680 priced at an assumed $90 | 5.5, 5.8, 6.3, 7.1 | record-from-docs arithmetic; priced, not saved |
| Seven definitions | Frozen before the first review page | 5.5, 6.3, 7.2 | record |
| Crux, raw and normalized | 25 against 63; about 27 against 25 on the same containers, sources, week, horizon; arrival variance 16 days against 8; 90 against an 80 north star after sources | 5.5, 5.7, 6.3, 7.1, 7.2 | record endpoints; populations illustrative |
| Crux base and dates | 79 customers; cancellations August 1, September 1, October 24, 2022; E-Team frame February 1, 2022 | 5.7, 7.1 | record-from-docs |
| Wave hold | Arrival accuracy 76 against 80 | 5.4, 7.1 | record |
| Roadmap | Month 18 to month 6; 66 percent less effort; about $5 million avoided once | 5.3, 7.1 | record; the model under it is modeled; avoidance, not annual |
| FFA scope | 905.9 engineer-months; 75.5 engineer-years; P0 14 percent; +15 percent testing; import launch April 2022, 11 partners, 10 countries | 5.6, 7.1 | record-from-docs |

| | | | |
|---|---|---|---|
| Austria | About 55 percent true to about 99 percent within a quarter; about $500,000 a year; about $8 million across 25 sites | 5.3, 7.1 | record, executed; $5 million base is a user anchor |
| Carrier bid | About 6 weeks to about 3; gate under 5 percent | not spoken here | user anchor; no award dollar |
| Available-for-sale | Signed June 5, 2025; oversell recurring to zero | 5.4, 5.6, 7.1 | record for the outcome; no baseline count |
| Architecture council | March 29, April 30, May 15, July 2, July 31, August 14; go-lives January and May 2025 | 5.6, 7.1 | record |
| Utah log and efficiency | 714 of 836 = 85.4 percent; efficiency +15 percent; zero customer incidents | 5.8, 7.1 | record; the 15 points have no before/after rates |
| Robotics gate | 98 percent bar; week two 94 percent; about 30 percent held back; full in week three | 5.4, 7.1, 7.2 | record |
| Cadence | Six to four weeks; defects down 50 percent | 5.7, 7.1 | record |
| Walmart promise; Google Fiber spine | No production figure | 7.1 | design case; illustrative only |

## 8.4 Public facts cited

Node names and network shape in this packet are drawn from these sources. Years are the year of the fact, with the publication year where it differs. Nothing in the worked examples is an Amazon figure.

| FACT | SOURCE |
|---|---|
| SLAM expanded as scan, label, apply, manifest. Path described as fulfillment center, then sort center, then delivery station. Before the first-party network, deliveries were completed by carriers such as USPS and UPS. Lanes are air and ground. A lane is one or more scheduled trucks, filled so the truck does not leave empty and the dock is not left holding packages. Amazon Science, October 14, 2022. | https://www.amazon.science/blog/maximizing-the-efficiency-of-amazons-own-delivery-networks |
| Fulfillment-center flow, including the SLAM step between pack and the outbound trailer, and the handoff to a sort center. Amazon fulfillment-center tour. | https://www.aboutamazon.com/news/operations/amazon-fulfillment-center-photo-tour |
| Middle mile from fulfillment centers by freight and air into sort centers, sorted by ZIP, then to delivery stations or carrier partners such as the local post office. Relay described as the tool carriers use to book Amazon loads. | https://www.aboutamazon.com/news/operations/powering-amazons-transportation-network |
| Critical pull time, in a 2016 MIT thesis on Amazon transshipment: the time an item must be picked, palletized, and on a truck. Order critical pull time: the time the package must depart Amazon, set from the ship-method cycle time. | http://hdl.handle.net/1721.1/106075 |

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| Delivery Estimate Accuracy, in a 2024 MIT thesis: the percentage of items with a delivery attempt on or before the promised delivery date. The thesis discusses virtual-physical mismatch and missort as causes. No rate is published. | https://hdl.handle.net/1721.1/155603 |
| Eight interconnected, largely self-sufficient U.S. regions, replacing one national fulfillment network. Andy Jassy, 2022 letter to shareholders, published 2023. | https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2022-letter-to-shareholders |
| In-region fulfillment rose from 62 percent to 76 percent. Pilot regions Northeast and Mid-Atlantic switched on January 18, 2023. Eight was the highest region count that preserved speed without excessive inter-regional inventory movement. Direct fulfillment-center-to-delivery-station trucks became easier to fill. Amazon Science, July 24, 2023. | https://www.amazon.science/news-and-features/how-amazon-reworked-its-fulfillment-network-to-meet-customer-demand |
| Full regional deployment by March 2023. Region design as ZIP-code clustering under demand and distance limits, then fulfillment-center assignment, then simulation, judged on cost to serve and delivery speed. Middle-mile buildings specialized into hubs, for long-haul fill, and last-mile sort centers, for delivery stations. INFORMS Journal on Applied Analytics, 2025. | https://pubsonline.informs.org/doi/full/10.1287/inte.2025.0295 |
| More than 76 percent of fulfilled orders from within the customer's region; distance to the customer down 15 percent since the start of 2023; middle-mile touchpoints down 12 percent; same-day buildings fulfill, sort, and deliver from one site; pick to outbound dock averages 11 minutes. Doug Herrington, July 31, 2023. | https://www.aboutamazon.com/news/operations/doug-herrington-amazon-prime-delivery-speed |
| In-region shipping to delivery stations avoided nearly 16 million miles in 2023. More than two-thirds of Amazon packages in 2023 delivered by Amazon's own logistics network. Truck fill named as an operating lever. 2023 Amazon Sustainability Report, published 2024. | https://sustainability.aboutamazon.com/2023-amazon-sustainability-report.pdf |
| U.S. cost to serve down by more than $0.45 per unit year over year in 2023; 58 same-day facilities. Andy Jassy, 2023 letter to shareholders, published 2024. | https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2023-letter-to-shareholders |
| 2024: second year of lower global cost to serve per unit; fourth-quarter transportation productivity from better inventory placement, higher units per package, and shorter distances. CFO remarks, Q4 2024 earnings call, February 6, 2025. | https://www.freightwaves.com/news/amazon-beats-q4-expectations-forecasts-slower-q1-growth |
| Rural plan: more than a billion packages a year across 13,000 ZIP codes and 1.2 million square miles. Andy Jassy, 2024 letter to shareholders, published 2025. | https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2024-letter-to-shareholders |

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| 2025: fastest-ever Prime speeds and lower cost to serve for the third year; next productivity levers named as distance, touches per package, and package consolidation. Q4 2025 earnings call, February 5, 2026. | https://www.fool.com/earnings/call-transcripts/2026/02/05/amazon-amzn-q4-2025-earnings-call-transcript/ |
| Same Day Fulfillment Centers, SSDs: more than 85 in the U.S.; more than $4 billion committed to the rural delivery network. Andy Jassy, 2025 letter to shareholders, published 2026. | https://www.aboutamazon.com/news/company-news/amazon-ceo-andy-jassy-2025-letter-to-shareholders |
| Delivery Service Partner program: independent businesses delivering for Amazon; launched 2018. | https://www.aboutamazon.com/impact/empowerment/delivery-partners |
| Last mile described as delivery stations plus Delivery Service Partners, Amazon Flex, and Hub Delivery. | https://www.aboutamazon.com/news/transportation/amazon-investment-delivery-network-small-town-rural-us |
| Amazon Flex: drivers use their own vehicles; Amazon.com blocks are typically 3 to 6 hours; pickup is at a delivery station. | https://flex.amazon.com/ and https://flex.amazon.com/support/faq |
| Delivery dates are shipping date plus transit time. An "order within" countdown is shown when it applies. Amazon Customer Service. | https://www.amazon.com/gp/help/customer/display.html?nodeId=GV8H5D3MMAR7JBLF |
| Amazon relies on a limited number of shipping companies to deliver inventory inbound and completed orders to customers. Form 10-K for the year ended December 31, 2025. | https://www.sec.gov/Archives/edgar/data/1018724/000101872426000004/amzn-20251231.htm |
| NielsenIQ click-to-door of 1.9 days for Amazon versus 4.4 days for other U.S. digital retailers, measured July 2022. eMarketer, January 11, 2023. | https://www.emarketer.com/content/delivery-sped-up-us-retailers-amazon |

### Looked for, and not used as an Amazon fact

An official definition or numeric standard for critical pull time: the term is used as the 2016 thesis defines it and the hours are the reference network. A published delivery estimate accuracy rate: none; the 97.7 percent is the reference network. Any published cost per package, trailer fill, sorter rate, route cost, or linehaul rate: none; every such number here is an input on the 8.2 card. A current mix of packages among Delivery Service Partners, Flex, USPS, and UPS: not published beyond "more than two-thirds by Amazon's own network" in 2023. Any internal system name for the transportation or network-design stack: not used, so none is recited as inside knowledge.

> **SAY THIS — 86 WORDS**
>
> Every number I used today is one of three things: a public Amazon fact with its year, a career number with its status, or an input to a reference network I built so the arithmetic could be shown. I will always say which. The public record gives the node names, the regional shift, and three years of lower cost to serve with faster speeds; it does not give a pull-time standard, a delivery-accuracy rate, or a cost per package, and I would not pretend it does.